6.1 Uniform Circular Motion

6.2 Centripetal Force, F_c

Learning Outcomes

6.1 UNIFORM CIRCULAR MOTION

a) **Describe** uniform circular motion

b) **Convert** units between degrees, radian and revolution (or rotation)

PREVIOUS KNOWLEDGE

An object moves in straight line if the net force on it, acts in the direction of motion OR the net force is zero – Chapter 4, Newton's law of motion

If the net force acts at an angle to the direction of motion ata any moment, the object moves in curved path. – Chapter 2, Projectile motion

6.1 Introduction

What is circular motion?

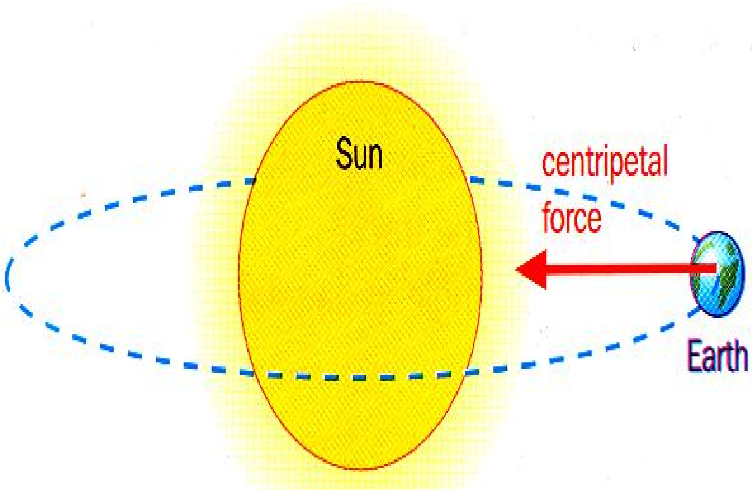
Circular motion – motion which occurs when bodies rotate around something or move in a circular path.

Example :

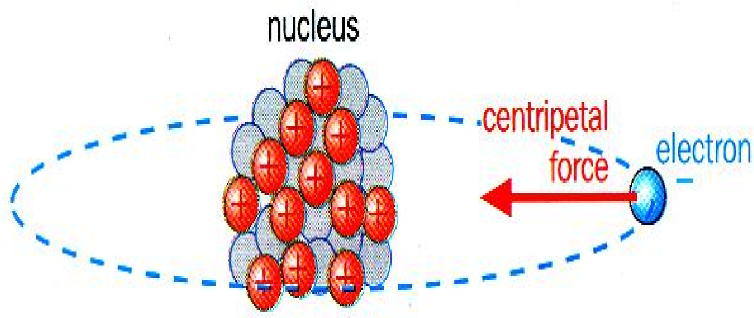
Roller Coaster cars

Eye on Malaysia

CD player



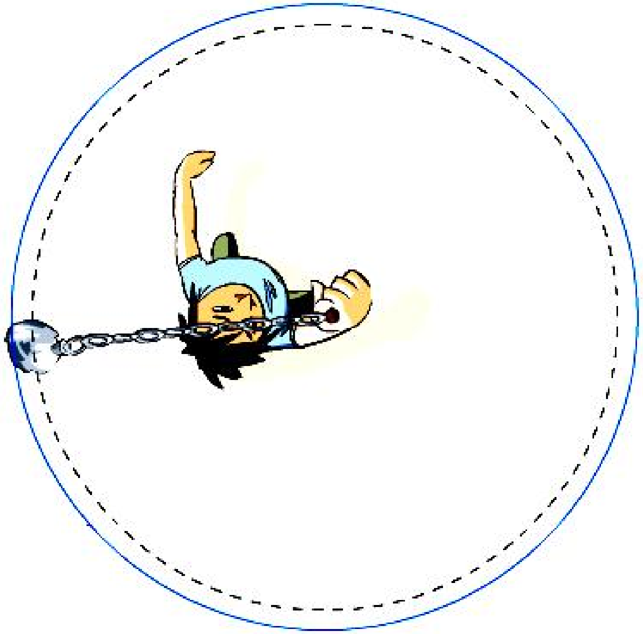
A planet orbiting Sun



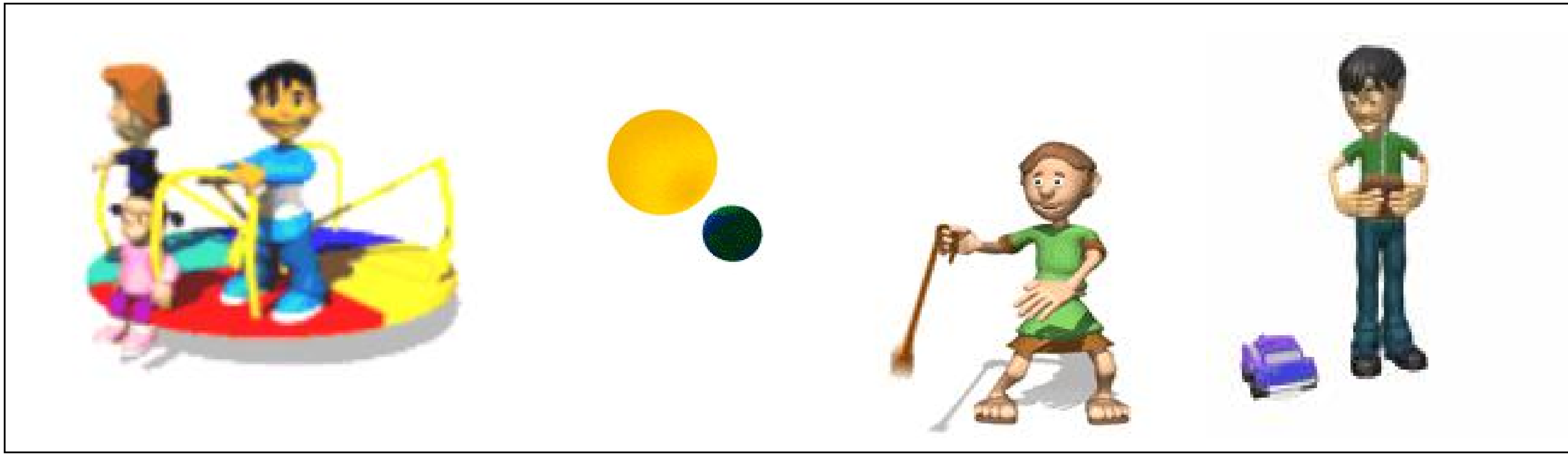
Circular Motion

Uniform

Non Uniform



6.1 a) Describe Uniform Circular Motion

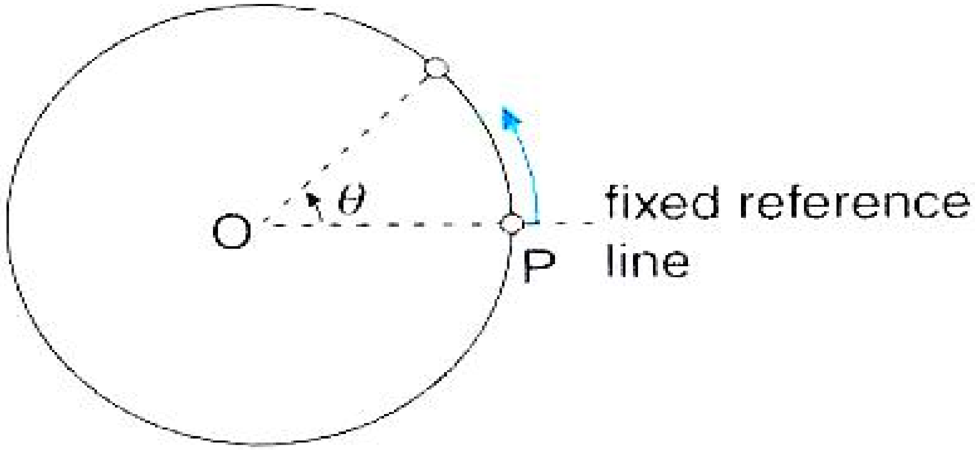


Uniform circular motion is defined as
the motion of an object in a circle
with a constant speed
(ie. magnitude of the velocity remains constant)
but the direction of the velocity
is continually changing with time.

Motion Characteristics for Circular Motion

- (A) Angular Displacement, θ
- (B) Linear Distance, s
- (C) Angular Velocity, ω
- (D) Linear Velocity, v
- (E) Frequency, f
- (F) Period, T

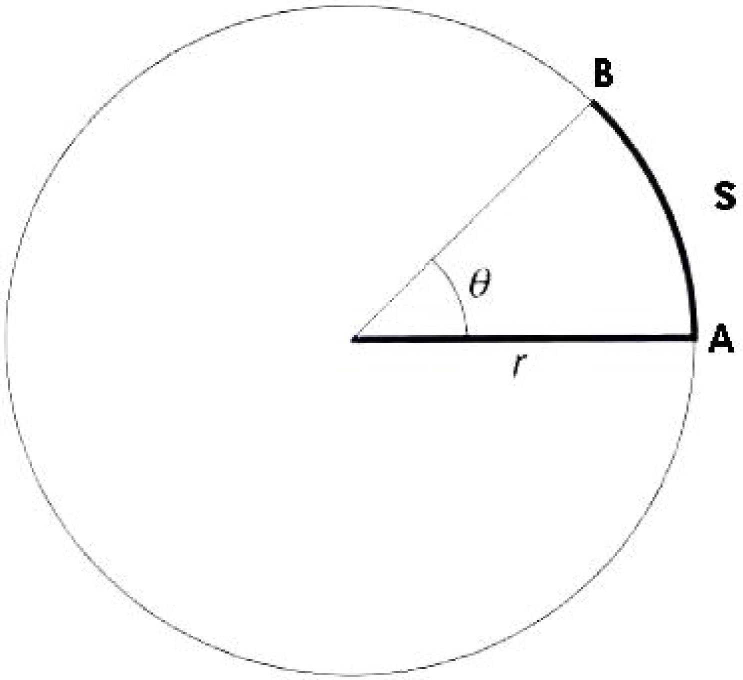
(A) Angular Displacement, θ



- The angle undergone by an object from a fixed reference point.
- Unit : radian (rad)
- Note :

$$\pi = 180°$$
$$2\pi = 360°$$

(B) Linear Distance, *s*



- When an object moves from point A to B in a circular path of radius r :

$$\theta = \frac{s}{r}$$

- rearranged the equation, we get :

$$\boxed{s = r\theta}$$

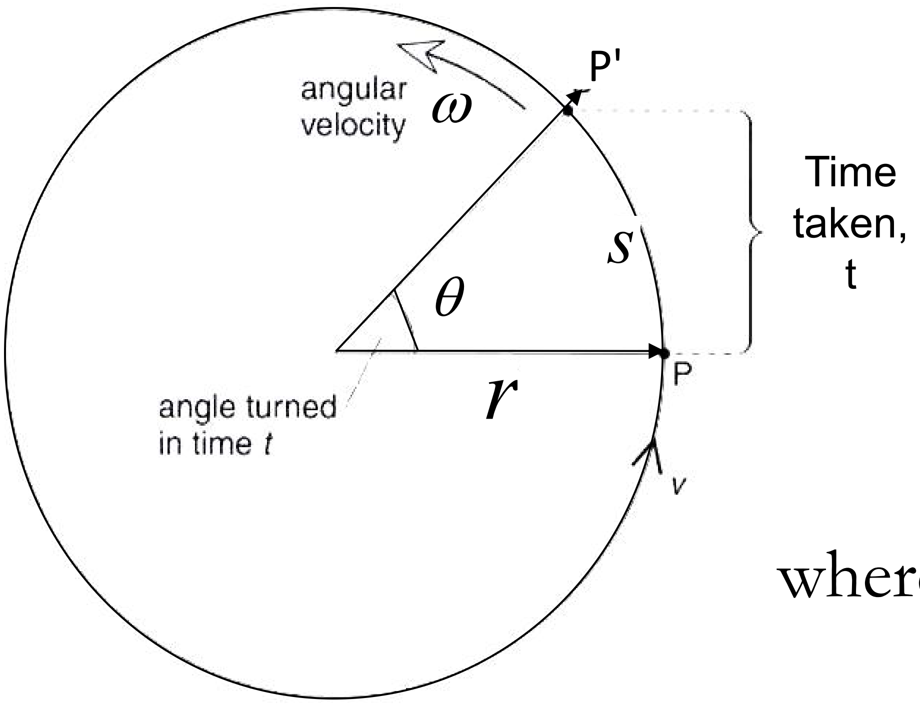
Where:

θ = angle turned through

r = radius of the circular path

Note: 360° = 2 π = 1 revolution

(C) Angular velocity, ω



- Defined as the rate of change of angular displacement

$$\boxed{\omega = \frac{\theta}{t}}$$

where θ : angle turned through
t : time interval

- Unit SI : rad s^{-1}
- Other units used : rps (revolutions per second) , rpm (revolutions per minute).

- Vector quantity

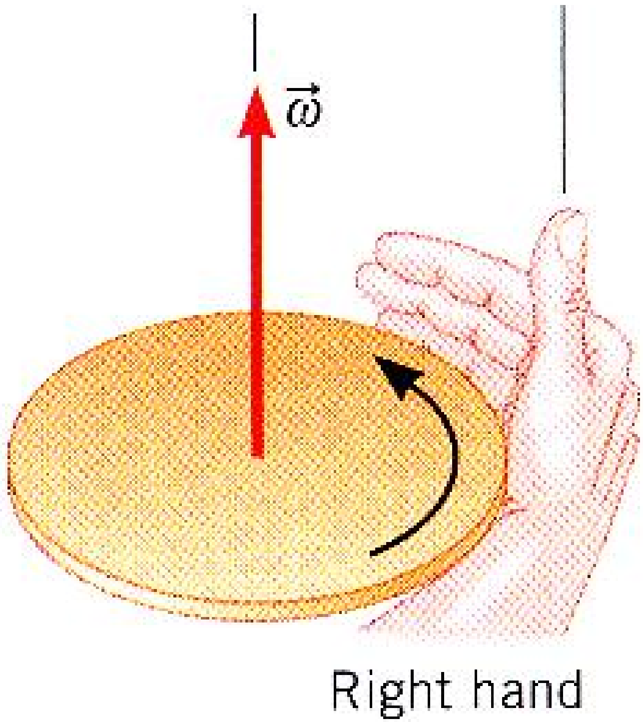
- also called ‘ angular frequency ’

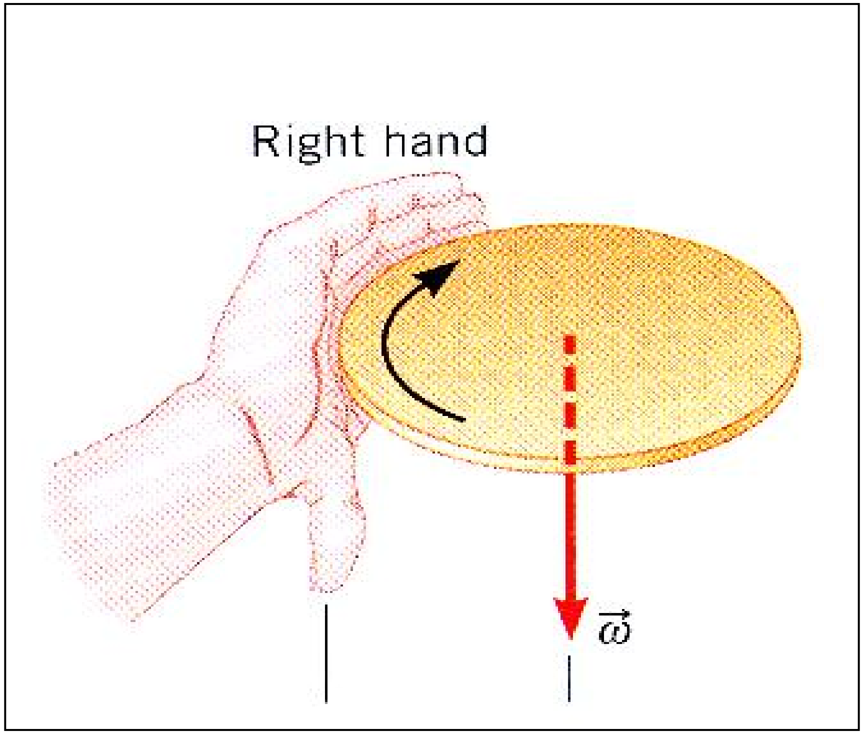
- Note : 1 rps = 2π rad s^{-1}

$$1 \text{ rpm} = \frac{2\pi}{60} rad\ s^{-1}$$

- The direction of the angular velocity, ω is depending on the rotation of the object (clockwise or anticlockwise).

When the fingers of the right hand curl around the circle and point the way it is rotate, the extended thumb points in the direction of $\vec{\omega}$





(D) Linear velocity, v

- An object moving in <u>uniform circular motion</u> would cover the same linear distance, s in each second of time, thus :

$$Linear\ speed, v = \frac{s}{t}$$

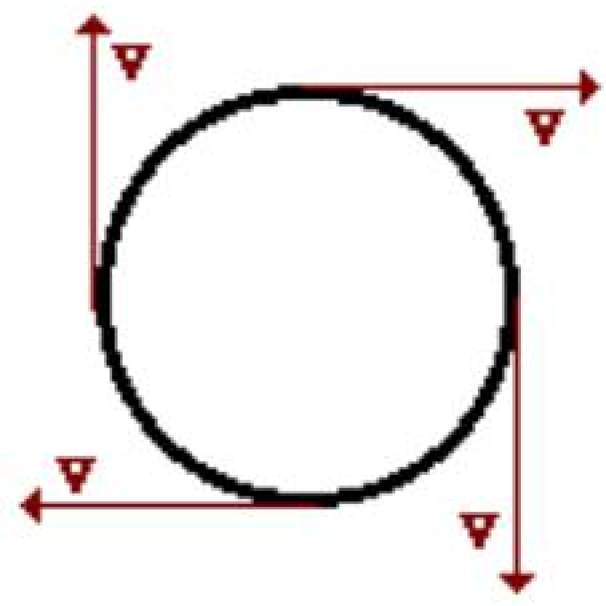
- Knowing that $s = r\theta$, substitute into above eq :

$$v = \frac{r\theta}{t} \Rightarrow \boxed{v = r\omega}$$

Where r : radius of the circle

ω : angular displacement

- The direction of linear velocity, v is always tangential to the circle which is perpendicular to the radius of the circle.



- In uniform circular motion, the magnitude of the linear velocity (speed) of an object is constant but the direction is continually changing.

- Unit SI : m s^{-1}

- also called tangential velocity

(E) Period, T

- time for an object to make one complete revolution.

- Unit SI : second (s)

(F) Frequency, f

- the number of revolutions performed per second.

$$f = \frac{1}{T}$$

- Unit : s^{-1} @ Hertz (Hz)

- One revolution = 2π radians. This takes T seconds.

From: $$\omega = \frac{\theta}{t}$$

$$\omega = \frac{2\pi}{T}$$

knowing that $f = 1/T$, thus → $\omega = 2\pi f$

Knowing, $v = r\omega$ substitute into above eq :

$$\boxed{v = \frac{2\pi r}{T} = 2\pi r f}$$

6.1 b) Convert units between degrees, radian and revolution (or rotation)

Degree	Radian	No of revolution (rev)
360°	2π radian	1 revolution

Example 1

Express the following angles in radian :

(a) 30° (b) 45°

Give as numerical values and as a fraction of π.

Solution

$$a)\ 30° = \frac{30° \times 2\pi}{360°} = \frac{\pi}{6}\ rad = 0.5236\ rad$$

$$b)\ 45° = \frac{45° \times 2\pi}{360°} = \frac{\pi}{4}\ rad = 0.7854\ rad$$

Express the following angles in radian :

(c) 60° (d) 180° (e) 425°

Give as numerical values and as a fraction of π.

Solution

$$c)\ 60^\circ = \frac{60^\circ \times 2\pi}{360^\circ} = \qquad rad = \qquad rad$$

$$d)\ 180^\circ = \frac{180^\circ \times 2\pi}{360^\circ} = \qquad rad = \qquad rad$$

$$e)\ 425^\circ = \frac{425^\circ \times 2\pi}{360^\circ} = \qquad rad = \qquad rad$$

$c)\ \frac{\pi}{3}\ rad = 1.0472\ rad\ ;\ \ d)\ \pi\ rad = 3.142\ rad\ ;\ \ e)\ 2.361\,\pi\ rad = 7.418\ rad$

Express 2 revolution in :

a) radian

b) degree

Solution

$$a)\ 2rev \times 2\pi = 4\pi\ rad$$

$$b)\ 2rev \times 360^\circ = 720^\circ$$

Example 2

An object undergoes circular motion with uniform angular speed 100 rpm. Determine :

(a) the period, T

(b) the frequency of revolution.

Solution

given : ω = 100 rpm

Convert to rad s^{-1} :

$$\omega = \frac{100\ rev}{1\,\mathrm{min}}$$

$$= \frac{100 \times 2\pi}{60s} = 10.47\ rad\ s^{-1}$$

(a) $$from\ \omega = \frac{2\pi}{T}; we\ get:$$

$$T = \frac{2\pi}{\omega}$$

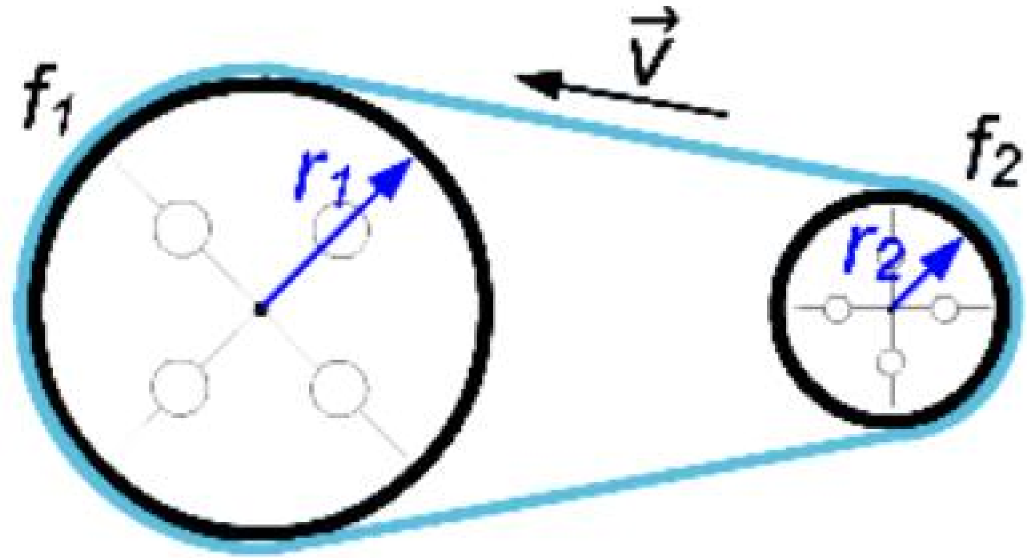
$$T = \frac{2\pi}{10.47} \quad \boxed{= 0.60 s}$$

(b) $$frequency, f = \frac{1}{T} = \frac{1}{0.6}$$

$$\boxed{= 1.67 Hz}$$

Example 3

2 wheels of a machine are connected by a transmission belt. The radius of the first wheel r_1 = 0.5 m, the radius of the second wheel r_2 = 0.125 m. The frequency of the bigger wheel equals 3.5 Hz. What is the frequency of the smaller wheel ?



Solution

Velocity of 1st wheel : $$v_1 = 2\pi r_1 f_1$$

Velocity of 2nd wheel : $$v_2 = 2\pi r_2 f_2$$

But $v_1 = v_2$, therefore :

$$2\pi r_1 f_1 = 2\pi r_2 f_2$$

$$\frac{f_2}{f_1} = \frac{r_1}{r_2}$$

$$f_2 = \frac{r_1}{r_2} \times f_1 \quad = \frac{0.5}{0.125} \times 3.5$$

$$\boxed{f_2 = 14\ Hz}$$

Example 4

An object travels around the circumference of a circle of radius 3 m at a rate of 15 rev min^{-1}. Calculate

(a) Its angular speed in rad s^{-1}.

(b) Its linear speed around the circular path.

Solution

(a) Given: $\omega = 15$ rev min^{-1} ; convert to rad s^{-1} :

$$\omega = \frac{15\,(2\pi)}{60} = \boxed{\frac{\pi}{2}\,rad\,s^{-1}}$$

(b) Using: $v = r\omega \quad = 3\left(\frac{\pi}{2}\right)$

$$\boxed{v = 4.71\,ms^{-1}}$$

Follow up Exercise

1. A particle is moving on a circular path of radius 0.5 m at a constant speed of 10 m s^{-1}. Find the time taken to complete 20 revolutions.

{ t = 6.28 s }

2. What are the angular velocities of the hour-hand and minute-hand of a clock? {1.45×10^{-4} rad s^{-1}, 1.75×10^{-3} rad s^{-1}}

3. The speed of an object in uniform circular motion is 6.0 m s^{-1}. the diameter of the circle is 2.5 m. What is it angular velocity?

{4.8 rad s^{-1}}

4. A particle moving in a circle takes 0.25 s to complete one revolution. How many complete revolutions per minute (rpm) does the particle perform? {240 rpm}

Follow up Exercise

5. A particle in uniform circular motion takes 3.0 s to move from one end A of a diameter to the other end B. The diameter of the circle is 5.00 cm. Calculate
   (a) the angular velocity?
   (b) the speed? { 1.05 rad s^{-1}, 2.62 x 10^{-2} m s^{-1} }

6. An object is moving in a circle of radius 40 cm with constant angular velocity of 6 rad s^{-1}. Find (a) the velocity of the object
   (b) the time for one complete revolution

   {2.4 m s^{-1}, 1.05 s}

7. A 200g of mass is moving with uniform angular velocity in a circle of radius 80 cm on a smooth horizontal surface. If the mass completes 10 revolutions in one second, calculate:
   (a) the angular velocity?
   (b) the linear velocity?

   { 62.8 rad s^{-1}, 50.3 m s^{-1} }

6.2 CENTRIPETAL FORCE

- **Define** centripetal acceleration.
- **Use** centripetal acceleration.

$$\boxed{a_c = \frac{v^2}{r}}$$

- **Define** centripetal force.
- **Use** centripetal force.

$$\boxed{F_c = \frac{mv^2}{r}}$$

- **Solve** problems related to centripetal force for uniform horizontal circular motion.

6.2 Centripetal Force, F_c

Centripetal Acceleration, a_c

An object moving in a circle of radius, r at a constant speed, v has an acceleration.

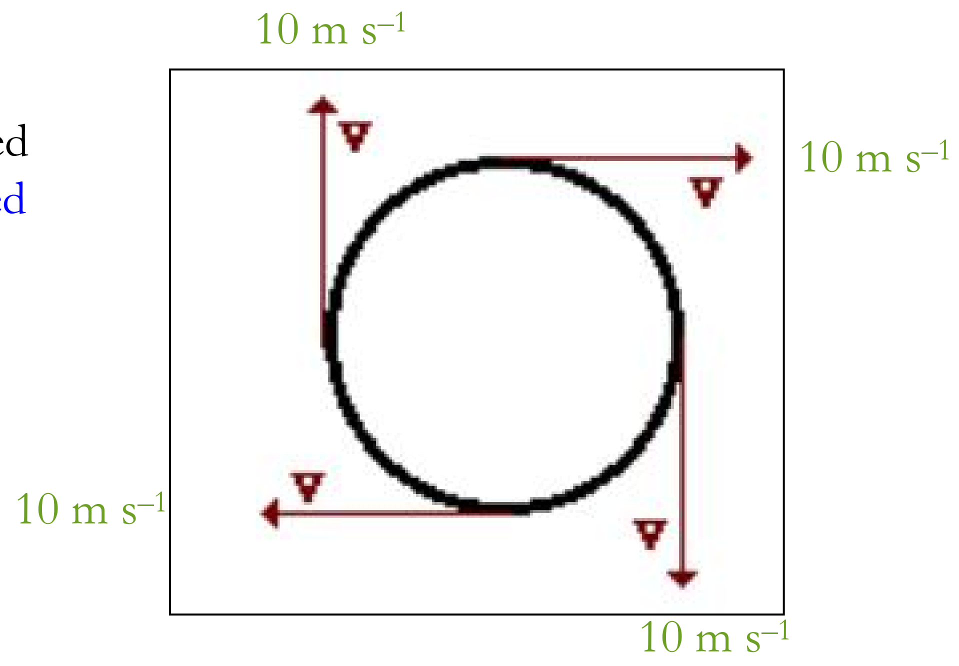
Why ?

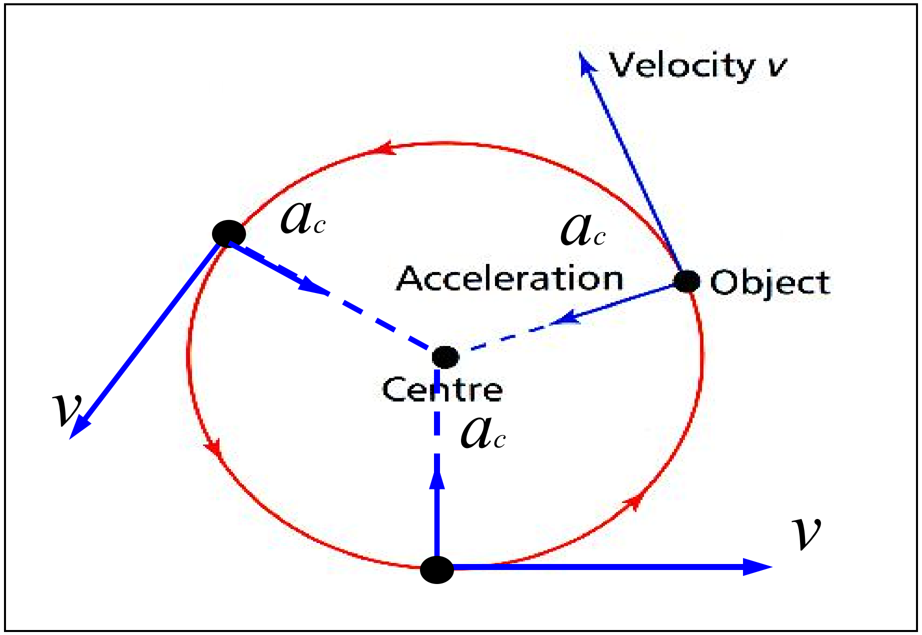
Although the object has constant speed but its direction of the tangential speed is continuously changing as the object turns the circle.

Velocity, being a vector quantity, has a constant magnitude but a changing direction → acceleration occurs.

This acceleration is called centripetal acceleration, a_c.

Centripetal means 'towards the center of the circle'.





Centripetal acceleration is defined as the **acceleration of an object moving in circular path** whose **direction is towards the centre** of the circular path and whose **magnitude is equal to the square of the speed divided by the radius**.

Direction of centripetal acceleration: always directed towards the center of the circle and **perpendicular to the linear(tangential) velocity**.

The magnitude of the a_c :

$$\boxed{a_c = \frac{v^2}{r}}$$

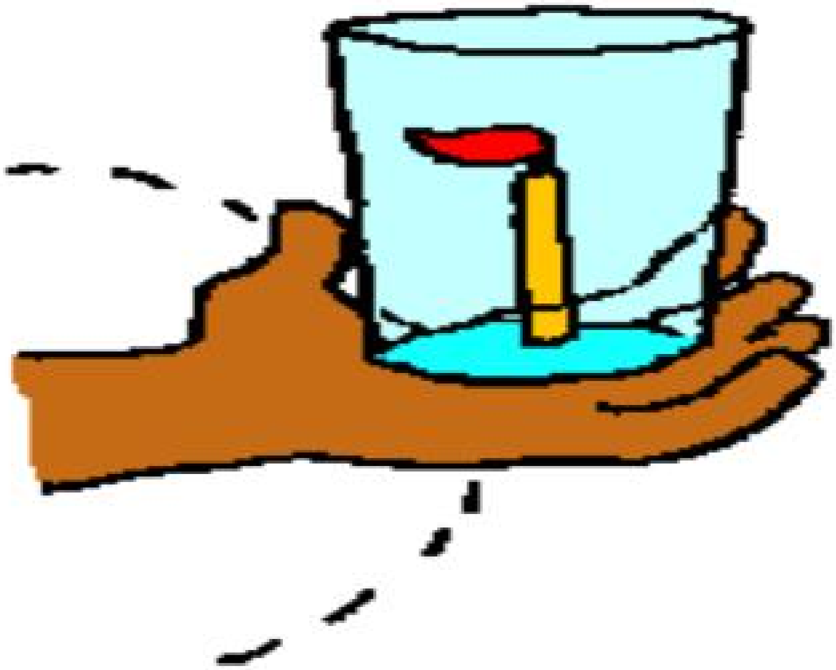
where v : speed of the moving object
r : radius of the circular path

since $v = r\omega$, thus we get :

$$\boxed{\begin{aligned} a_c &= r\omega^2 \\ a_c &= v\omega \end{aligned}}$$

Simple Experiment

Glass is held at the end of an outstretched arm as you spin in a circle at a constant speed.


A candle flame accelerometer indicates an inward acceleration

Example

Calculate the centripetal acceleration of a car traveling on a circular racetrack of 1000 m radius at a speed of 180 km h^{-1}.

Solution

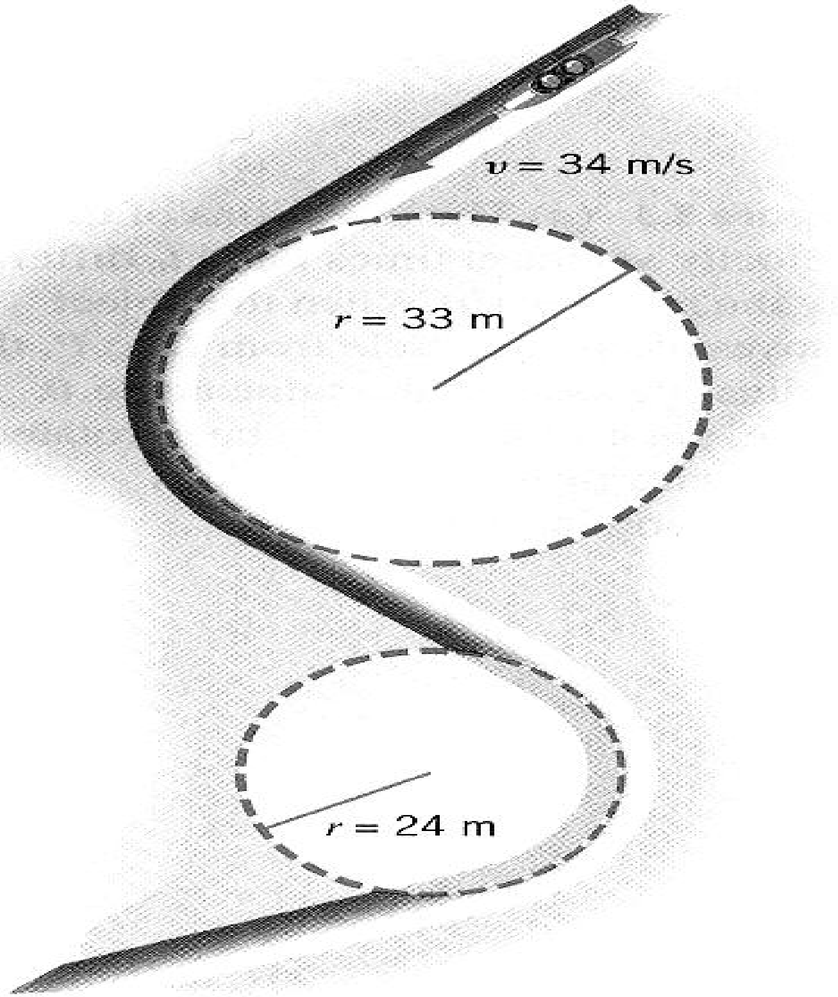
given : r = 1000 m
v = 180 km h^{-1}

$$= \frac{180(1000)m}{60(60)s} = 50 m\, s^{-1}$$

Using : $a_c = \frac{v^2}{r} = \frac{(50)^2}{1000}$ $\boxed{= 2.5 ms^{-2}}$

Follow Up Exercise

The bobsled track at the 1994 Olympics in Norway, contained turns with radii of 33 m and 24 m, as Figure below illustrates. Find the centripetal acceleration at each turn for a speed of 34 m s^{-1}.



Answer:
35.03 m s^{-2} ; 48.17 m s^{-2}

Centripetal Force, F_c

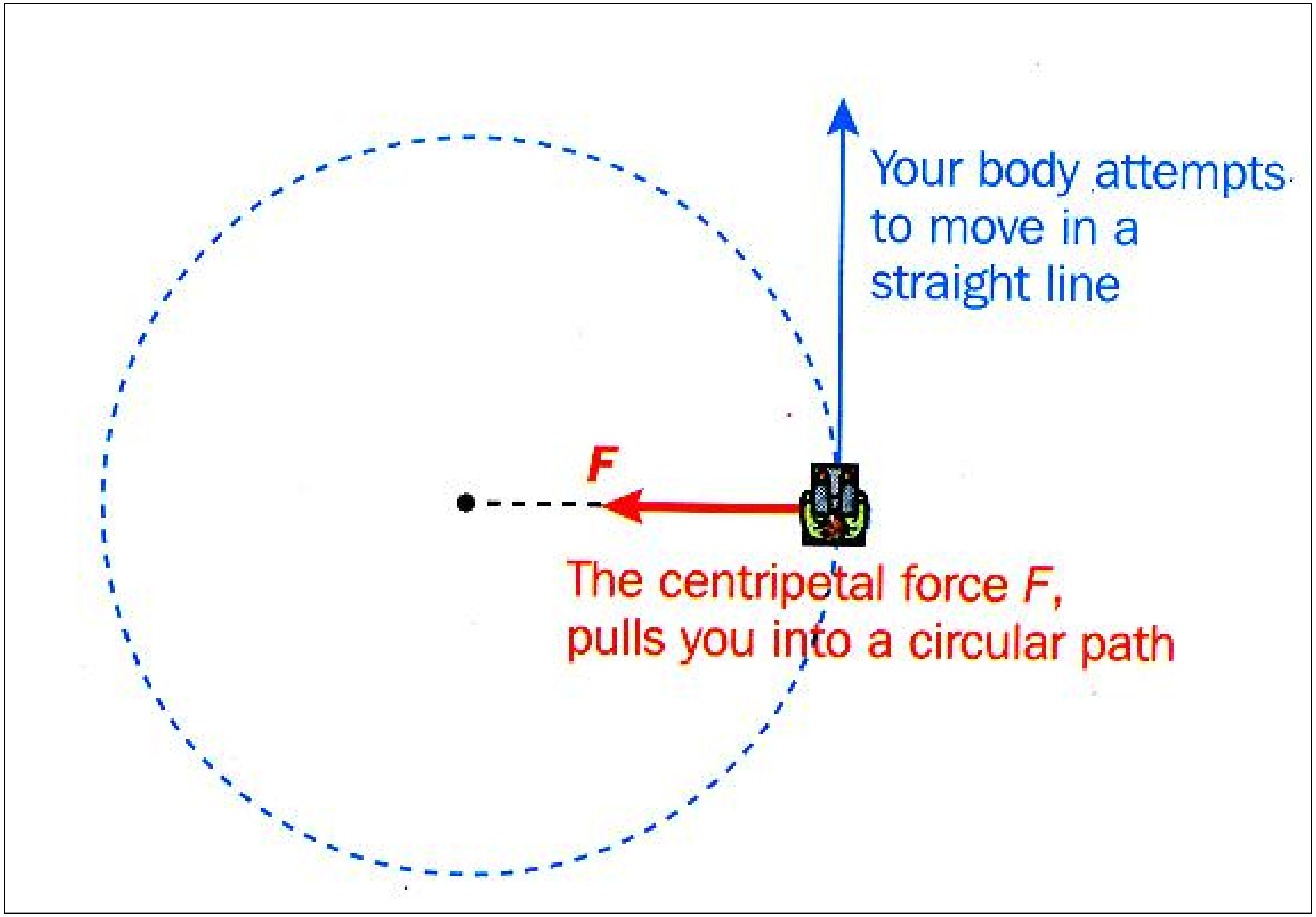
Newton's 2nd Law of motion indicates that :

" Whenever an object accelerates, there must be a net force to create the acceleration and the net force points in the same direction as the acceleration "

$$\sum \vec{F} = \vec{F}_{net} = m\vec{a}$$

Thus, for an object moving in a circle, there must be a net force acting upon it in order to cause its centripetal (or inward) acceleration.

This net force is known as Centripetal Force, F_c or Center seeking force.



Definition of Centripetal force

Centripetal force is the net force required to keep an object of mass m, moving at a speed v on a circular path of radius r.

Magnitude of F_c is given by :

$$F_c = ma_c$$

$$\boxed{F_c = \frac{mv^2}{r}}$$

where

m : mass of the object

v : speed of the object

r : radius of the circular path

Since v = r ω , thus :

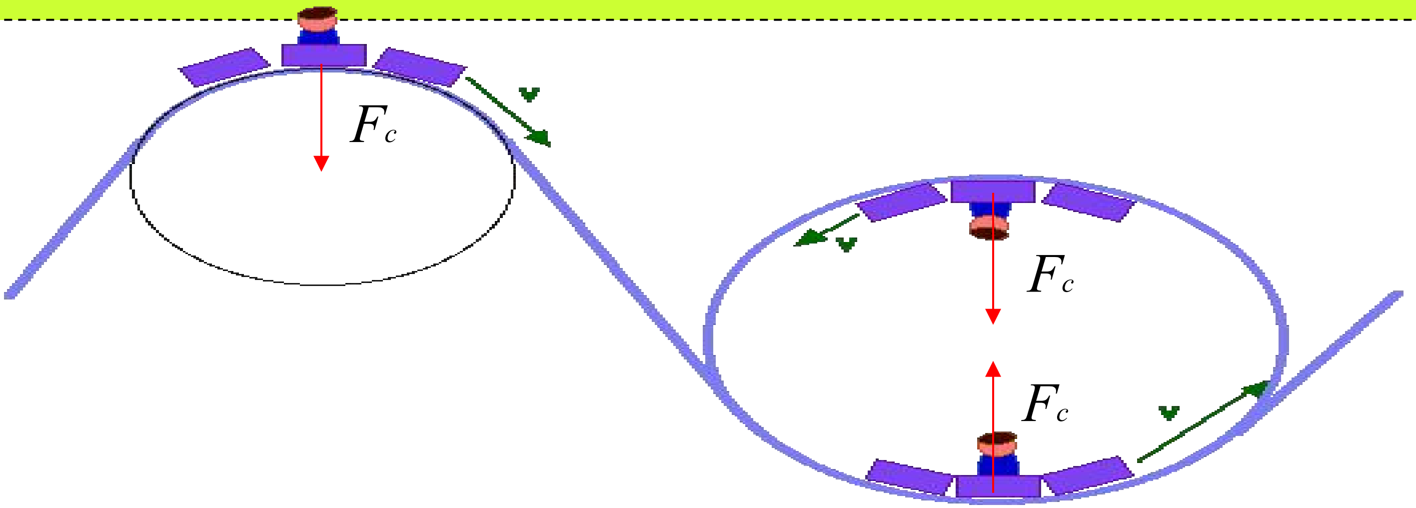
$$F_c = \frac{m(r\omega)^2}{r} \Rightarrow \boxed{F_c = mr\omega^2}$$

or $$\boxed{F_c = mv\omega}$$

Direction of F_c :

- Always act towards the centre of the circle.
- Same direction as the centripetal acceleration.

Where does the centripetal force come from ?



F_c can be a combination of forces such as tension in string, friction on the car's tires, normal reaction force or gravitational force.

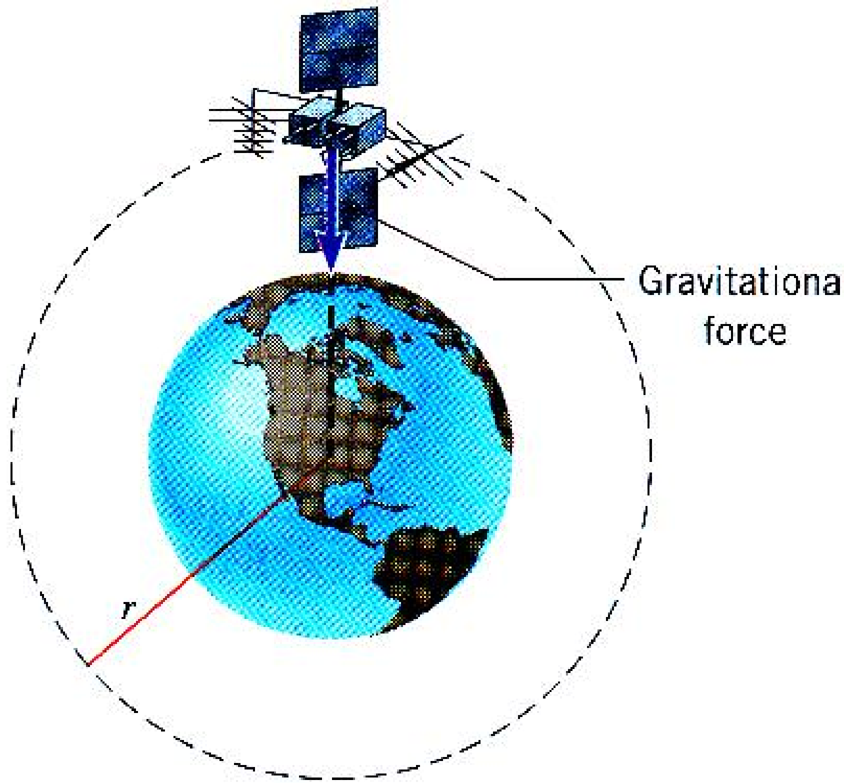
Centripetal force is NOT a NEW force acting on the object...

Centripetal force maintains the object's circular motion.

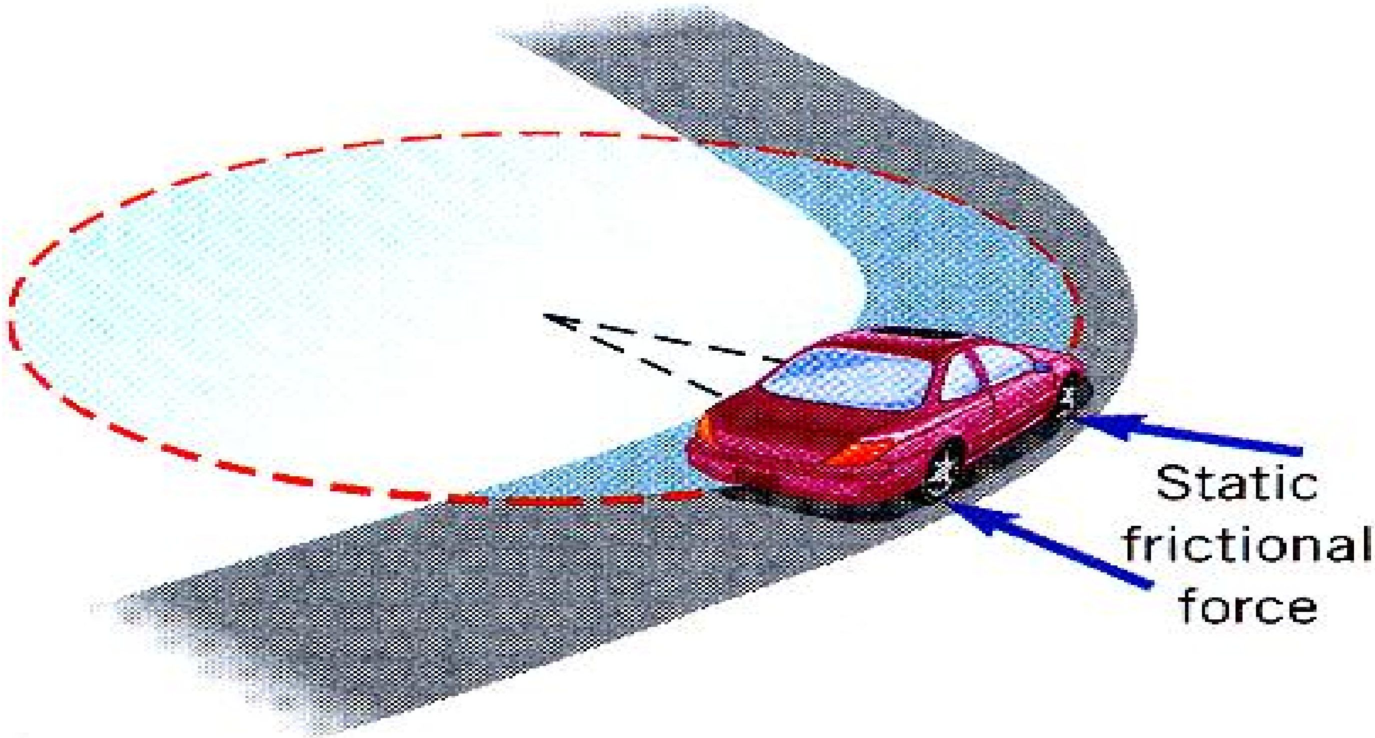
Centripetal force is just the name we give to the net force acting on the object in a direction towards the center of the circle.

In some case, it is easy to identify the source of the centripetal force.

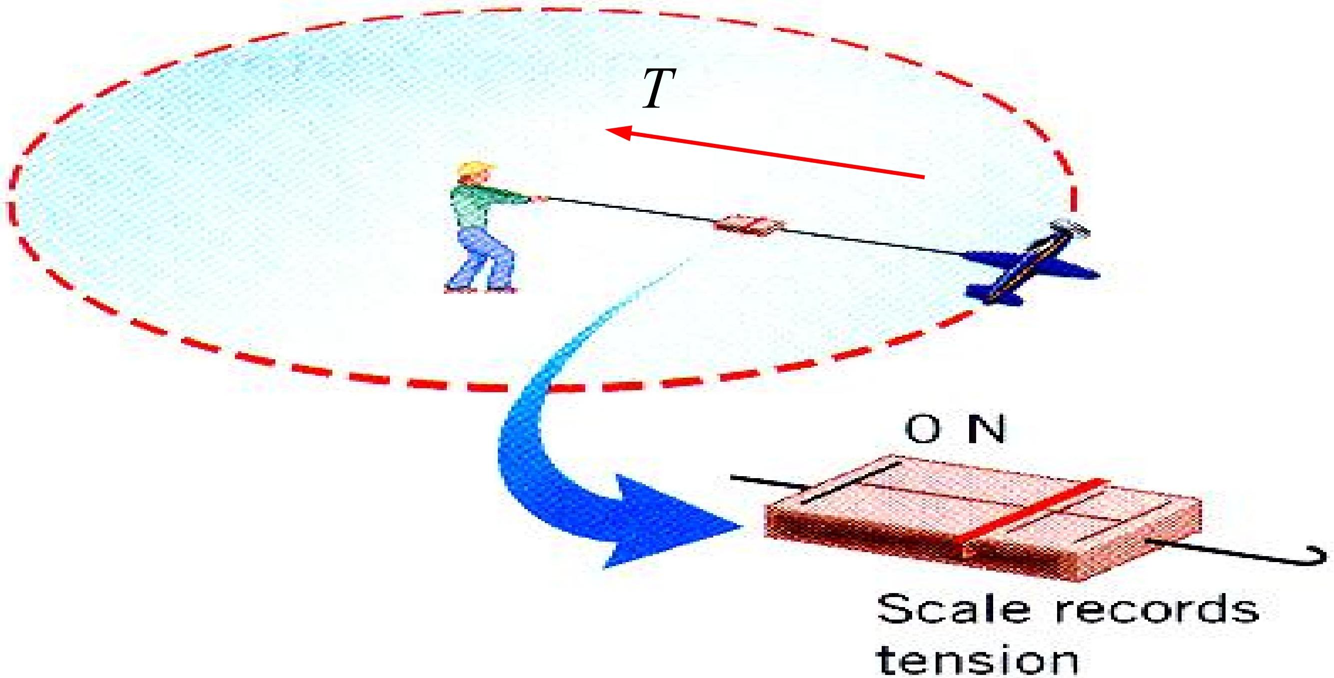
example



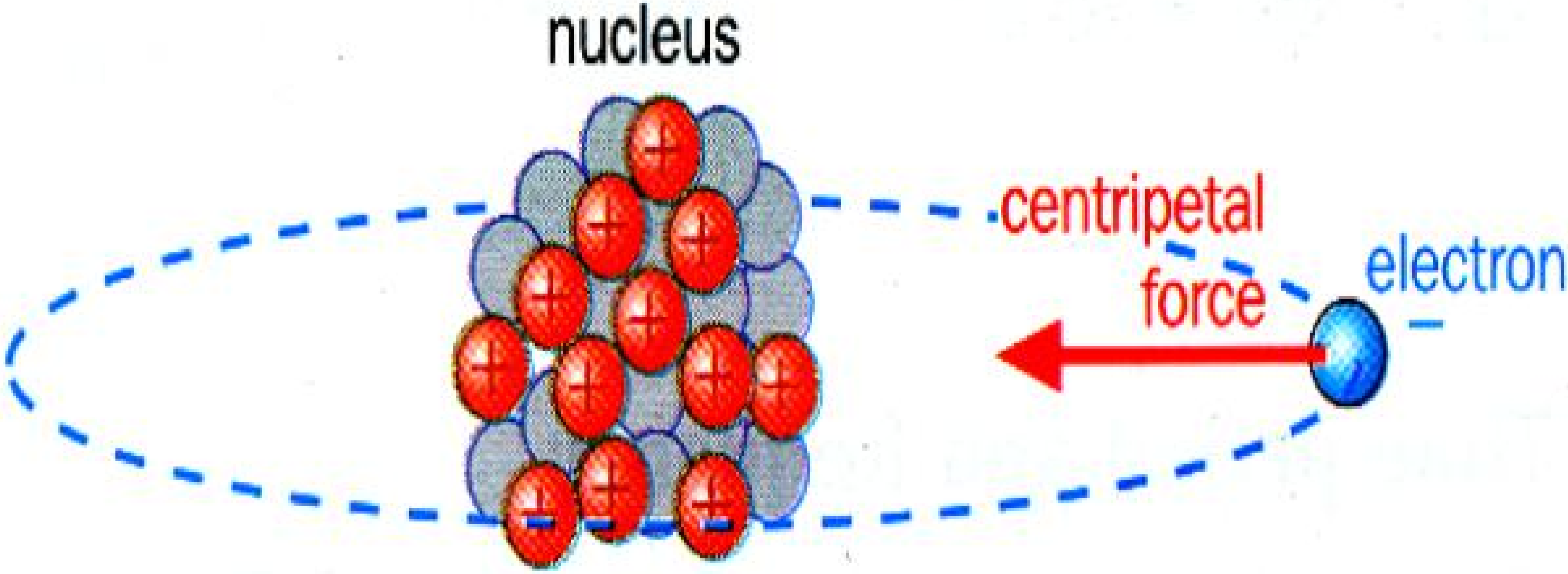
The earth's gravitational force on the satellite provides the centripetal force that keeps it in orbit.



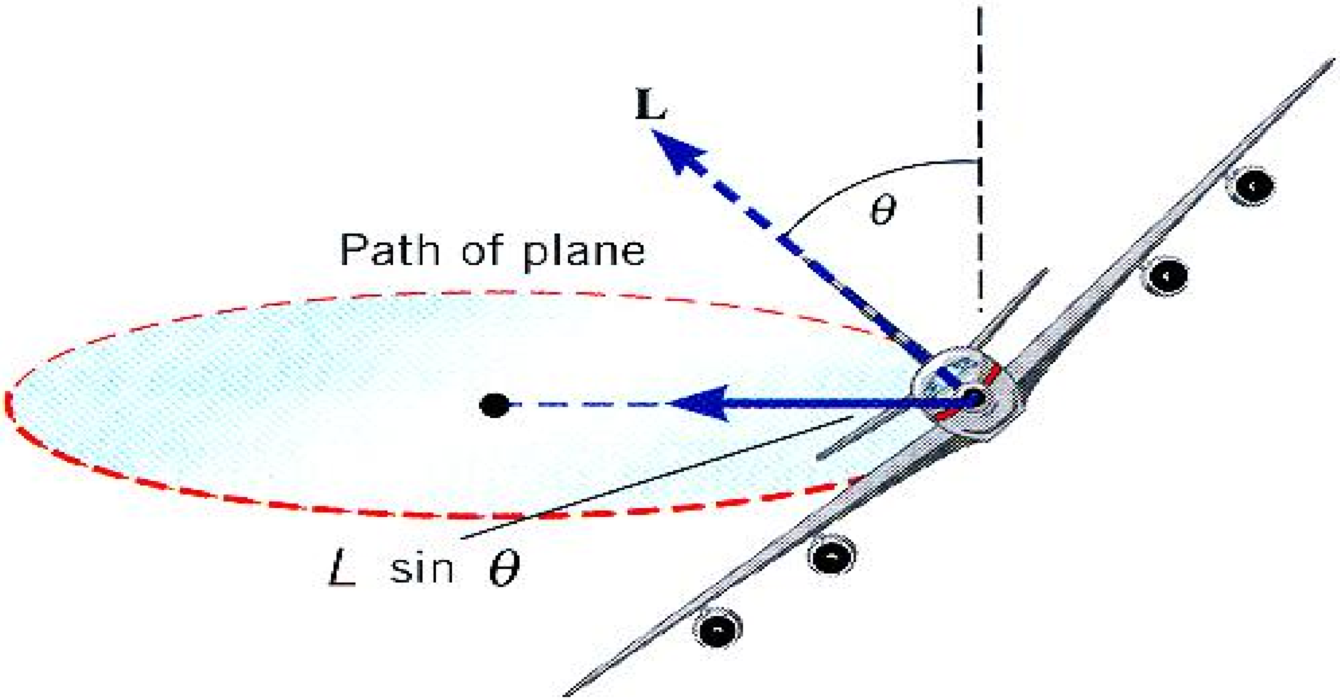
As a car makes a turn, the force of friction acting upon the turned wheels of the car provide the centripetal force that required for circular motion.



As when a airplane on a guideline flies in a horizontal circle, tension in the line directed toward the center of the circle provides the centripetal force.

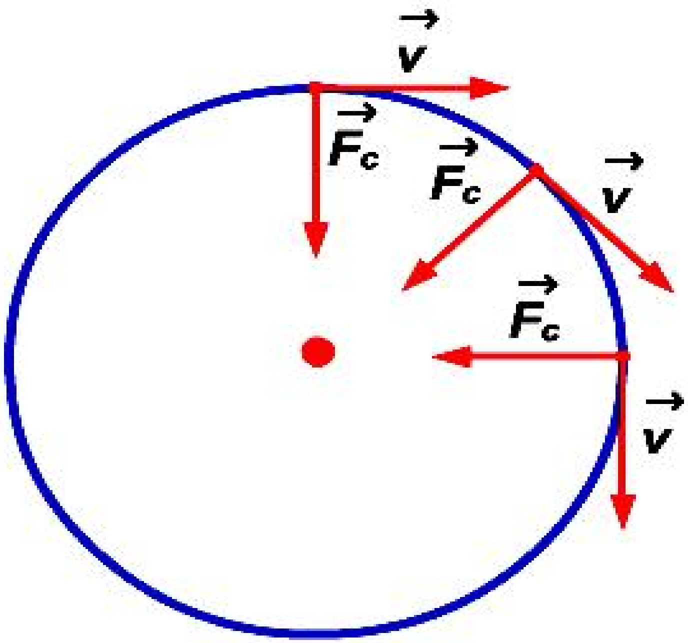


The attraction force between positively charged nucleus and negatively charged electron provided centripetal force.



When a plane executes a circular turn, the plane banks at angle θ. The lift component L sin θ is directed toward the center of the circle and provides the centripetal force.

F_c is at the perpendicular / right angle (90°) to the direction of the object's velocity.

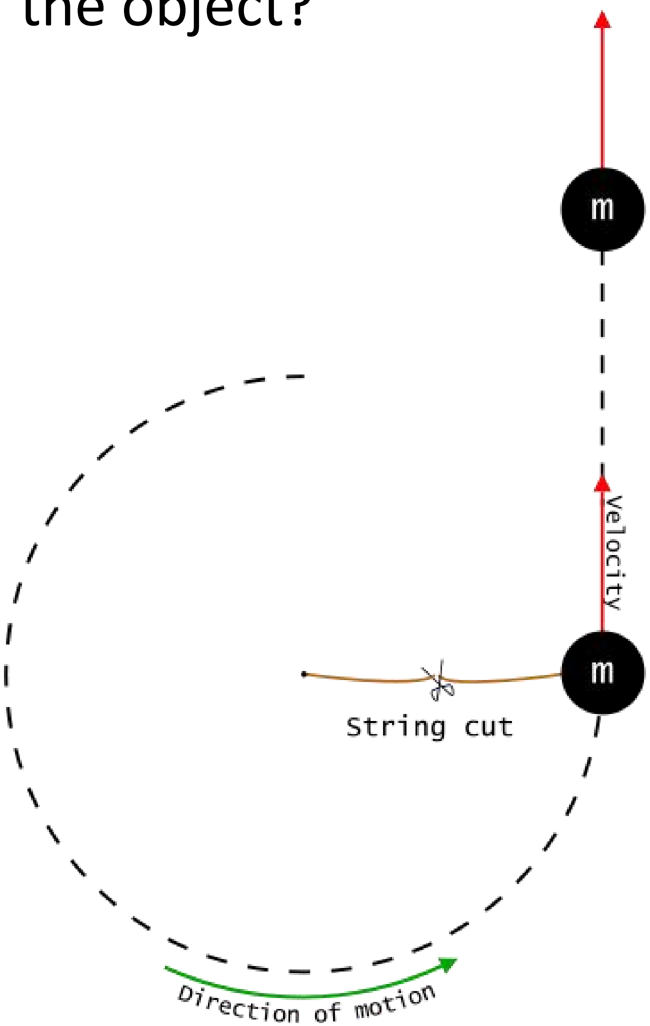


No work is done by the centripetal force, F_c

because

no displacement in the direction of the force.

What happens if the centripetal force suddenly stops acting on the object?



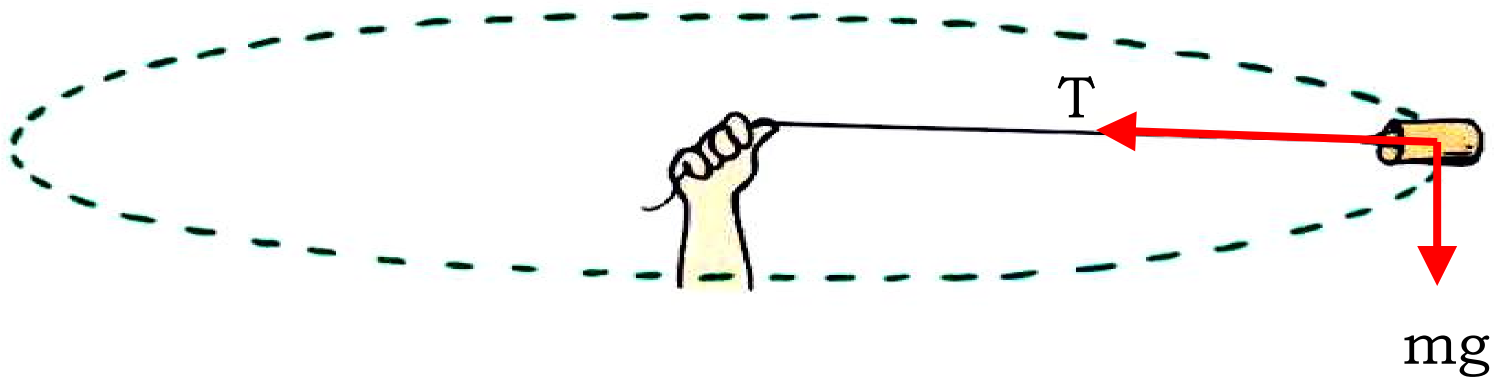
Like the diagram, if the string is being cut or snaps, no more centripetal force acts on the object.

The object will then continue its motion along a straight line in the direction of *v*, which is tangential to the circle.

- Uniform circular motion
  - Horizontal Circular Motion
    - Object revolving in a horizontal circle
    - Flat curve
    - Conical Pendulum
  - Vertical Circular Motion
    - Ferris Wheel
    - Objec evolving in a vertical circle

Uniform Circular Motion: Horizontal circle

CASE 1 : object revolving in a horizontal circle with steady speed.



2 forces acting on the object :

1. Force of gravity (weight, W = mg)
2. Tension force in the string, T

Tension in string, T is the only component in the radial direction that provides the centripetal force.

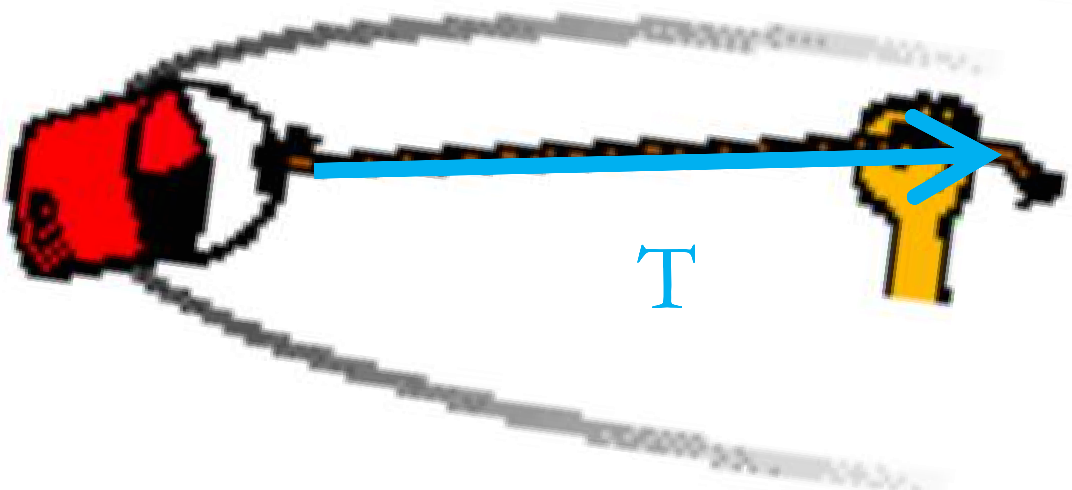
Applying Newton 2nd Law :

$$F_{net} = ma_c$$

$$F_{net} = F_c$$

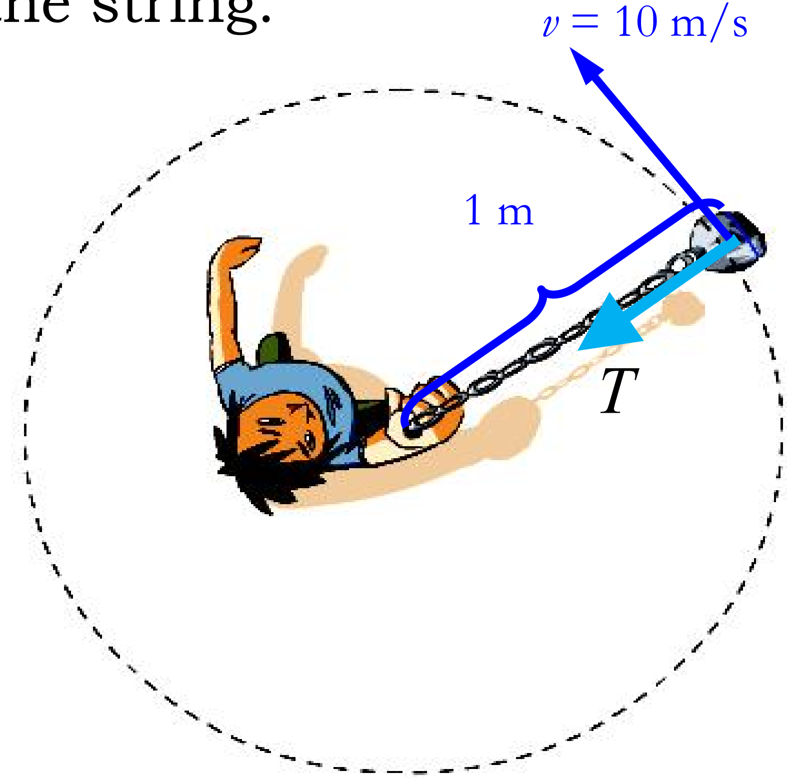
$$F_{net} = \frac{mv^2}{r}$$

$$\boxed{T = \frac{mv^2}{r}}$$



Example

A 0.25 kg rock attached to a string is whirled in a horizontal circle at a constant speed of 10.0 m s^{-1}. The length of the string is 1.0 m. Neglecting the effects of gravity, find the tension in the string.



Solution

$$T = F_c = \frac{mv^2}{r}$$

$$= \frac{0.25(10)^2}{1.0}$$

$$\boxed{T = 25\ \mathrm{N}}$$

Example

A ball of mass 150 g is attached to one end of a string 1.10 m long. The ball makes 2.00 revolution per second in a horizontal circle.

a. Sketch the free body diagram for the ball.

b. Determine

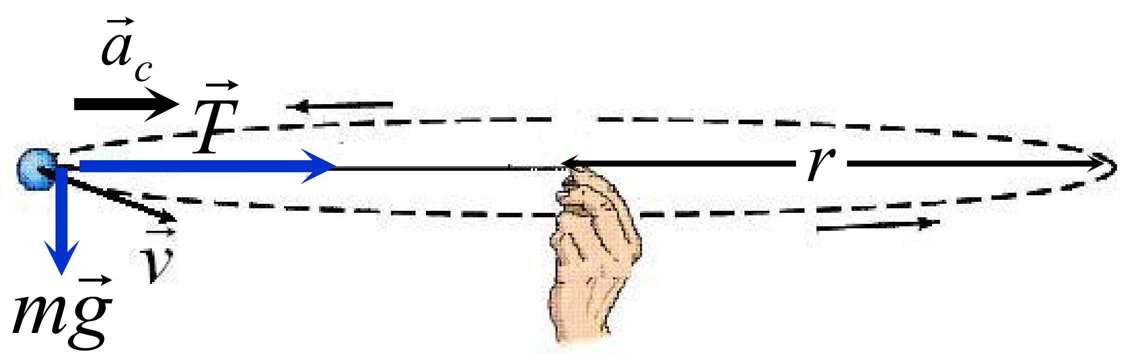
i. the centripetal acceleration of the ball,

ii. the magnitude of the tension in the string.

Solution :

$$m = 0.150\text{ kg}; l = r = 1.10\text{ m}; f = 2.00\text{ Hz}$$

a. The free body diagram for the ball :



Solution : $m=0.150\text{ kg};\ l=r=1.10\text{ m};\ f=2.00\text{ Hz}$

b. i. The linear speed of the ball is given by

$$v=\frac{2\pi r}{T}=2\pi rf$$

$$v=2\pi(1.10)(2.00) \Rightarrow v=13.8\text{ m s}^{-1}$$

Therefore the centripetal acceleration is

$$a_c=\frac{v^2}{r} \Rightarrow a_c=\frac{(13.8)^2}{1.10}$$

$$a_c=173\text{ m s}^{-2}$$

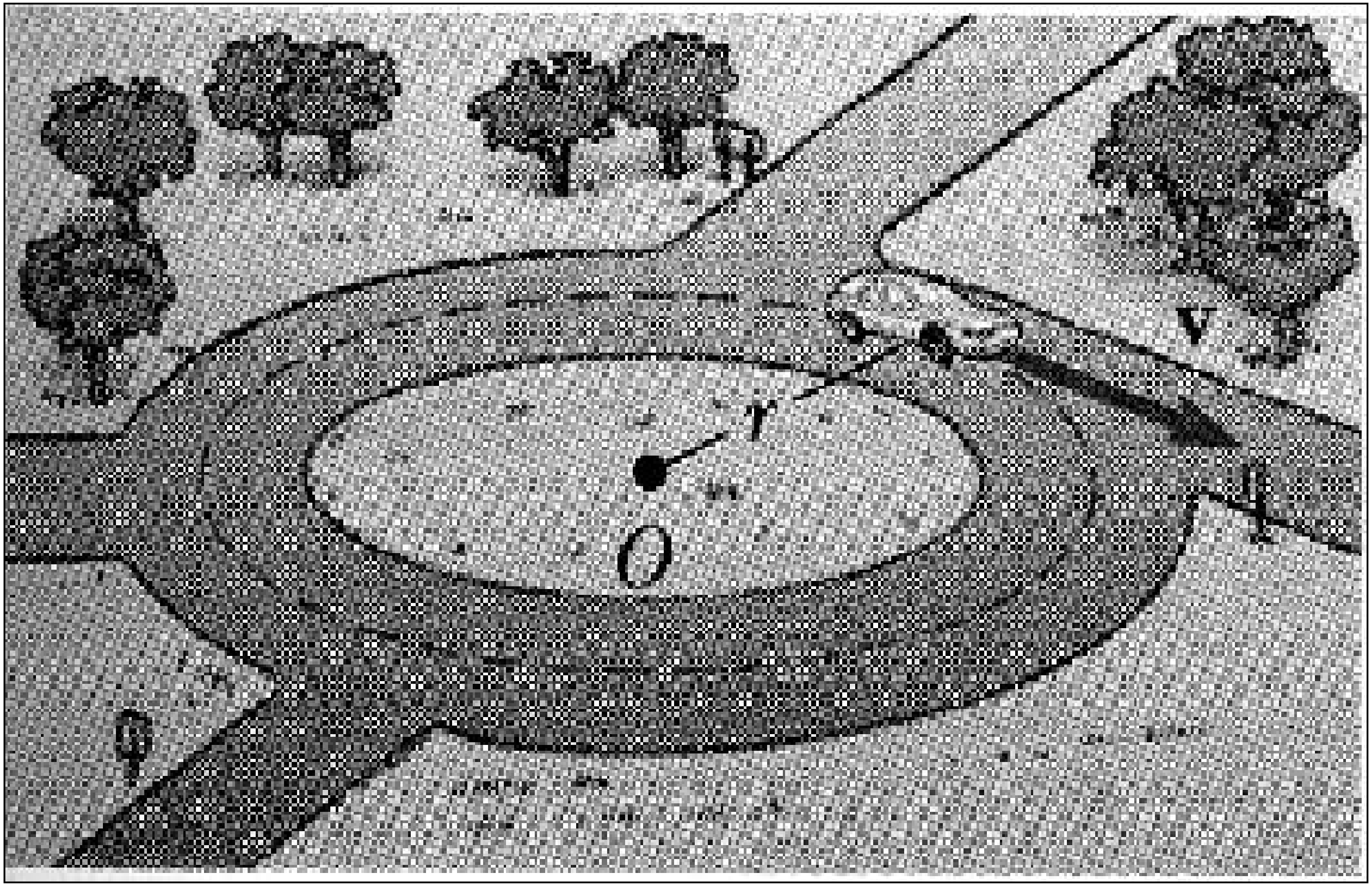
(towards the centre of the horizontal circle)

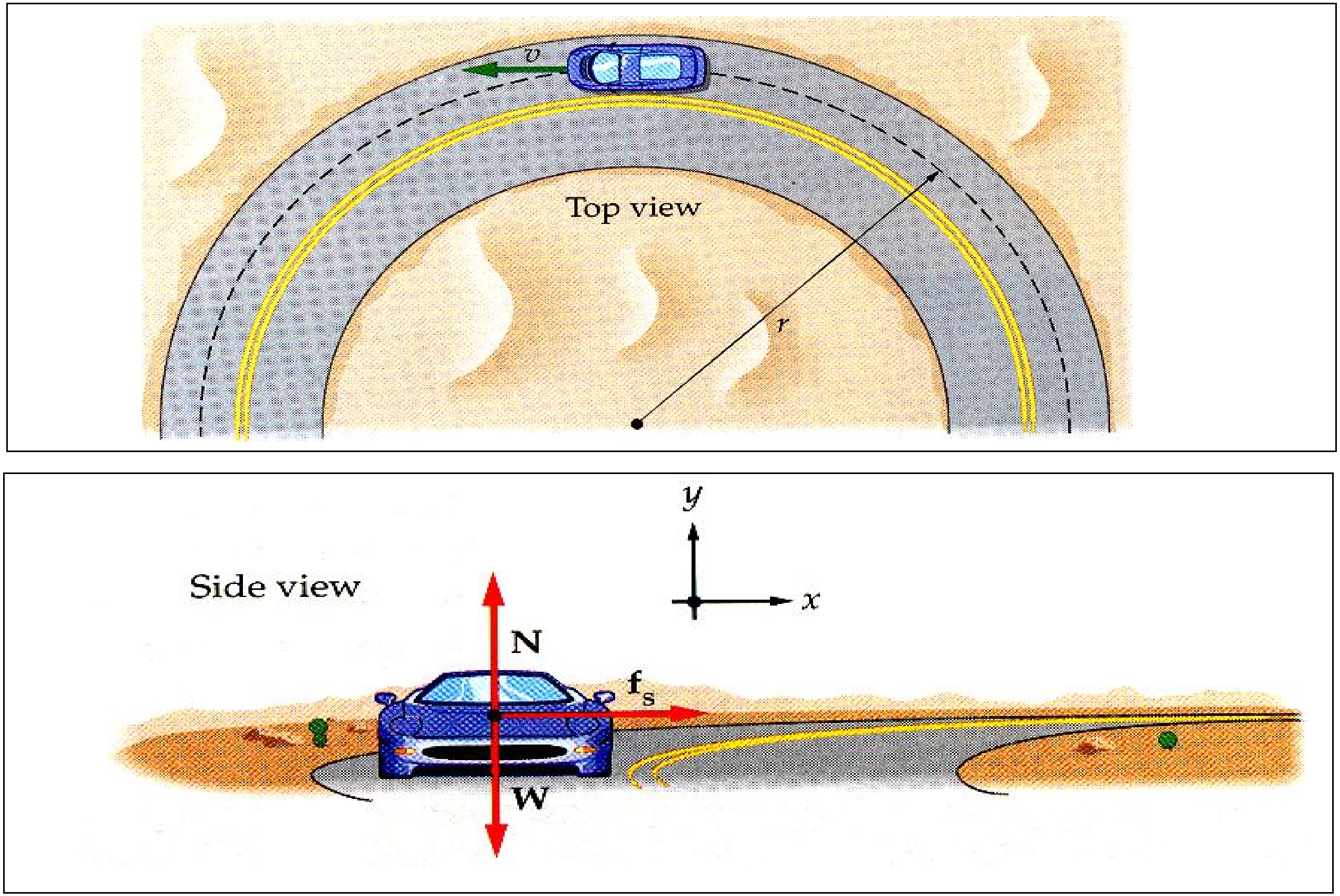
ii. From the diagram in (a), the centripetal force enables the ball to move in a circle is provided by the tension in the string.

Hence $$\sum F_x=F_c=ma_c$$

$$T=ma_c \Rightarrow T=(0.150)(173)=26.0\text{ N}$$

CASE 2 : Motion of car round a flat curve




v
Top view
r
y
x
Side view
N
f_s
W

<u>Refer to the horizontal component (x):</u>

Friction force, f_s between the tyres and the road, points toward the center of the circle, which provides the centripetal force.

$$F_{net} = F_c$$

$$\boxed{f_{s\,(\max)} = \frac{mv^2}{r}}$$

If the coefficient of static friction between the tyres & the road is μ_s :

$$\mu_s N = \frac{mv^2}{r} \qquad (\text{where } N = W = mg)$$

$$\mu_s (mg) = \frac{mv^2}{r}$$

The maximum velocity without slipping on the round curve is given by :

$$\boxed{v = \sqrt{\mu_s g r}}$$ (On a flat curve)

Let's try this example!

A car travels around an flat turn of radius $r = 50$ m. The coefficient of the static friction between the tyres and the road in dry weather is $\mu_s = 0.75$. Find the maximum speed at which the car can safely travel without skidding.

Example A car travels around an flat turn of radius r = 50 m. The coefficient of the static friction between the tyres and the road in dry weather is $\mu_s = 0.75$. Find the maximum speed at which the car can safely travel without skidding.

Solution

$f_{s(max)}$ supplies the centripetal force

$$f_{s(\max)} = \frac{mv^2}{r}$$

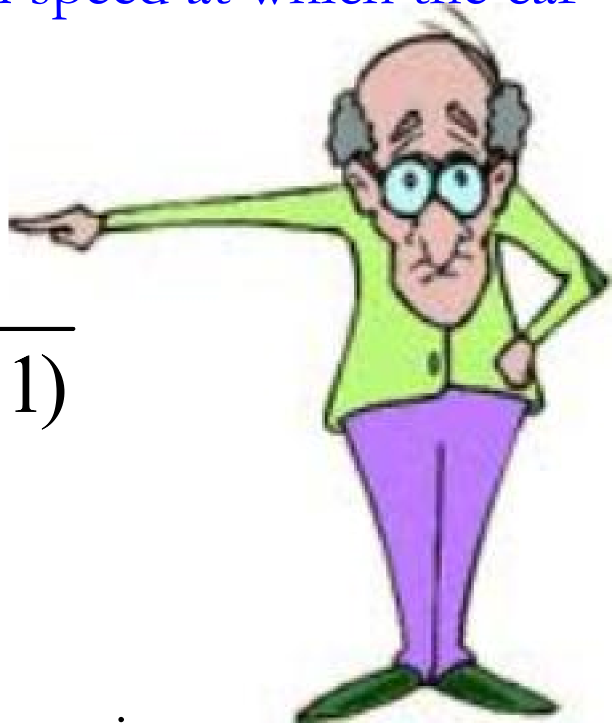
$$\mu_s m g = \frac{mv^2}{r}$$

$$v = \sqrt{\mu_s r g}$$

$$= \sqrt{(0.75)(50)(9.81)}$$

$$\boxed{v = 19.18 \text{ m s}^{-1}}$$

Refer to previous example, if given in icy weather, the coefficient of static friction = 0.11, calculate the maximum speed at which the car can safely negotiate the flat turn.



Solution

$$v = \sqrt{\mu_s r g} = \sqrt{(0.11)(50)(9.81)}$$

$$\boxed{v = 7.35 \text{ m s}^{-1}}$$

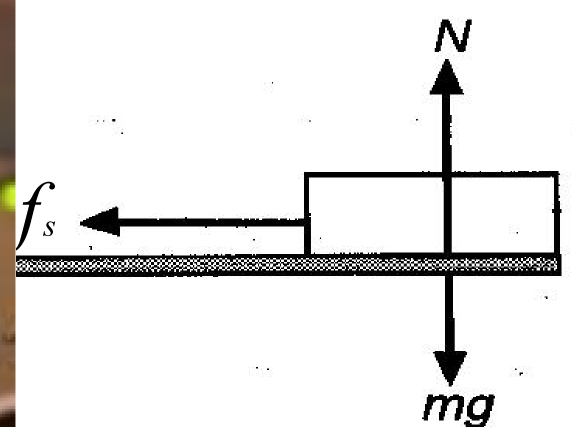
As expected, the dry road allows the greater maximum speed than the icy road.

Example

A 50 mg plastic puck is placed on the edge of a 30 cm radius phonograph record. The record is gradually brought to its usual speed of 33.3 rpm (rev min^{-1}). How large must the coefficient of static be if the puck is to stay on the record ?

Solution

Find : μ_s = ?



Given :

m = 50 mg = 50×10^{-3} g

r = 30 cm = 30×10^{-2} m

ω = 33.3 rpm

$$\omega = \frac{33.3(2\pi)}{60}$$

$$= 3.49\, rad\ s^{-1}$$

Assume coefficient of the static friction is μ_s

Refer figure :

$$f_s = F_c$$

$$\mu_s N = \frac{mv^2}{r} \quad \cdots(1)$$

component – y : $$N = mg \quad \cdots(2)$$

Substitute (2) into (1):

$$\mu_s(mg) = \frac{mv^2}{r}$$

$$\mu_s = \frac{v^2}{rg} = \frac{(r\omega)^2}{rg}$$

$$= \frac{[(30\times10^{-2})(3.49)]^2}{(30\times10^{-2})(9.81)}$$

$$\boxed{\mu_s = 0.37}$$

Example

A car of mass 2000 kg rounds a circular turn of radius 20 m. The road is flat and the coefficient of friction between tires and the road is 0.70.

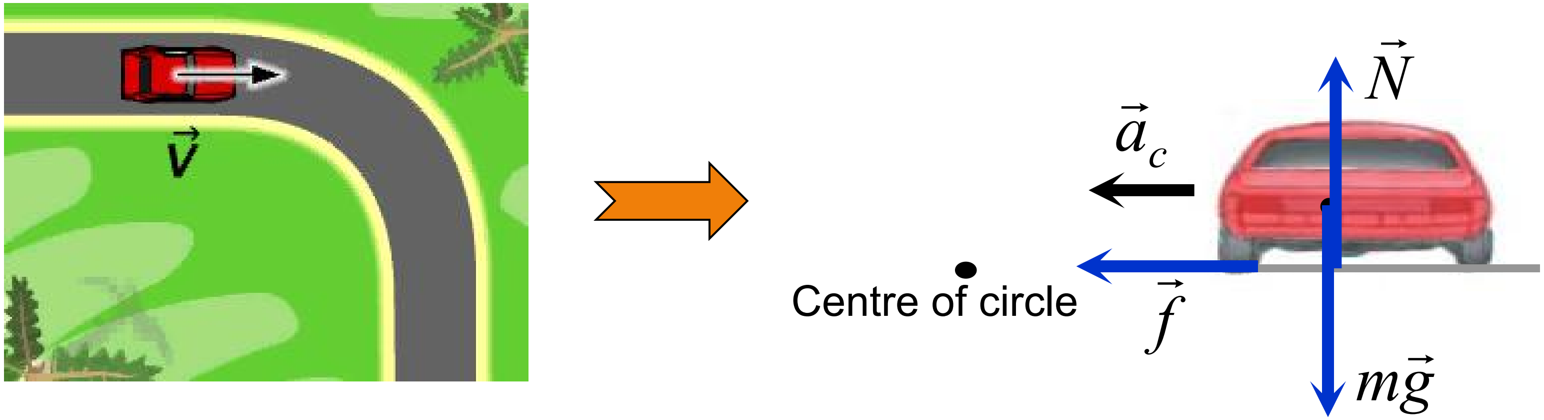
a. Sketch a free body diagram of the car.

b. Determine the maximum car's speed without skidding.

(Given g = 9.81 m s^{-2})

Solution : $m = 2000\ \text{kg};\ r = 20\ \text{m};\ \mu = 0.70$

a. The free body diagram of the car :



Solution : $m = 2000\ \text{kg};\ r = 20\ \text{m};\ \mu = 0.70$

b. From the diagram in (a),

y-component : $\sum F_y = 0 \Rightarrow N = mg$

x-component : The centripetal force is provided by the frictional force between the wheel (4 tyres) and the road. Therefore

$$\sum F_x = \frac{mv^2}{r}$$

$$f = \frac{mv^2}{r}$$

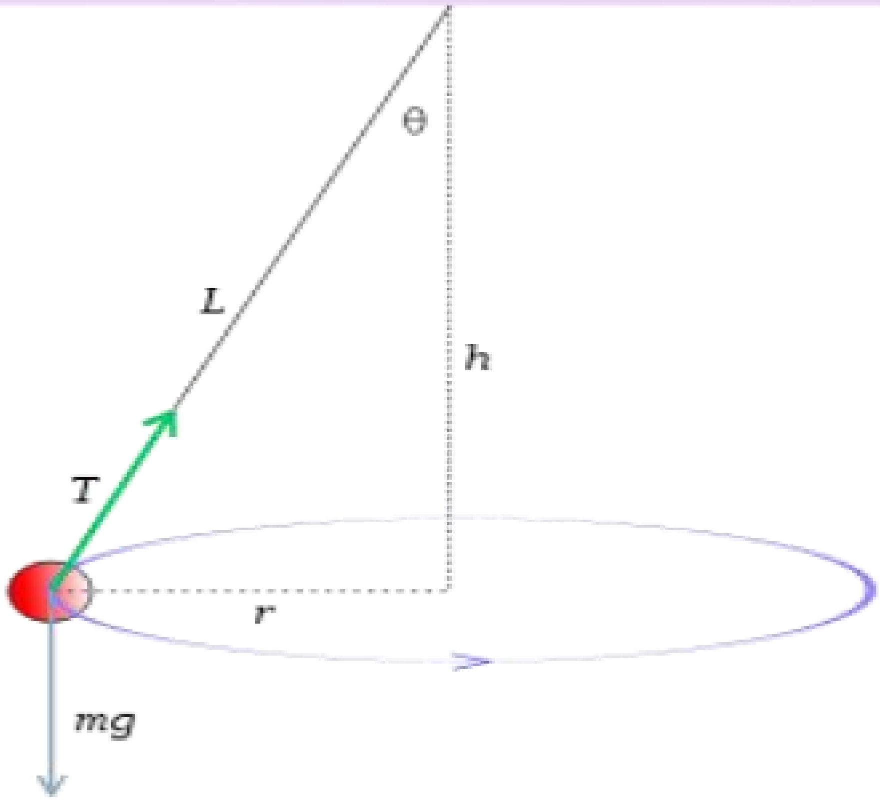
$$\mu mg = \frac{mv^2}{r}$$

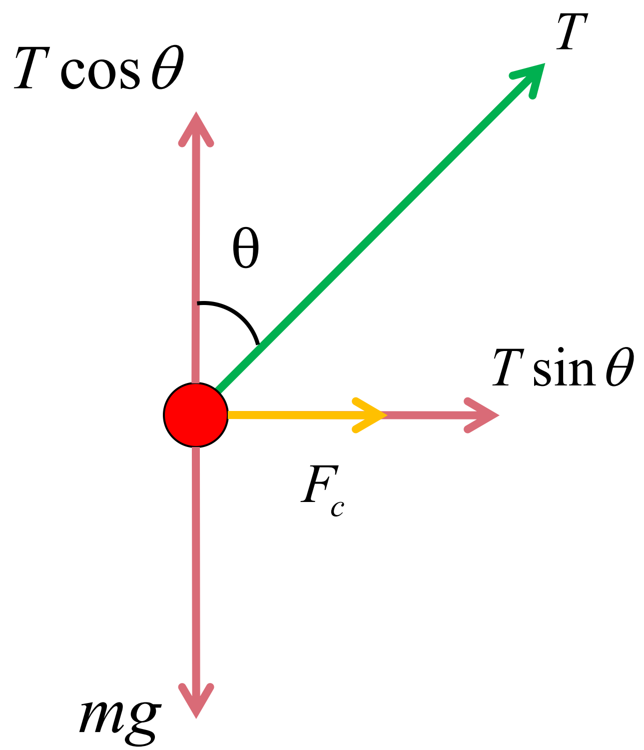
$$v = \sqrt{\mu rg}$$

$$v = \sqrt{(0.70)(20)(9.81)}$$

$$v = 11.7\ \text{m s}^{-1}$$

CASE 3 : Conical Pendulum





T sin θ° is the unbalanced force that supplies centripetal force.

component – x :

$$F_{net} = F_c$$

$$T\sin\theta = \frac{mv^2}{r} \quad \cdots(1)$$

component – y :

$$\Sigma F_y = 0$$

No acceleration for vertical component

$$T\cos\theta - mg = 0$$

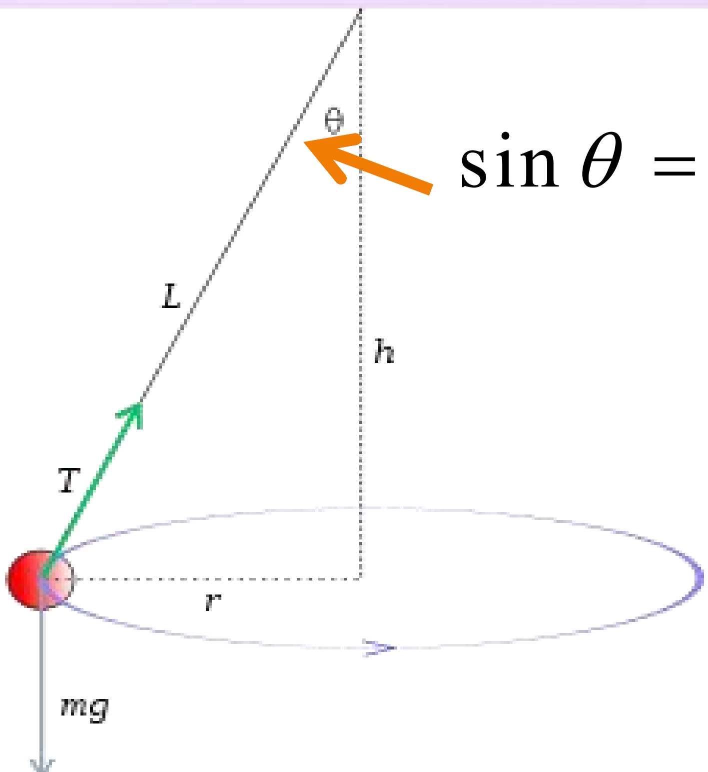
$$T\cos\theta = mg \quad \cdots(2)$$

$$\frac{(1)}{(2)}: \qquad \frac{\cancel{T}\sin\theta}{\cancel{T}\cos\theta} = \frac{\left(\frac{\cancel{m}v^2}{r}\right)}{\cancel{m}g}$$

$$\tan\theta = \frac{v^2}{rg}$$

$$\boxed{v = \sqrt{rg\tan\theta}}$$

From the diagram, we know : r = L sin θ :

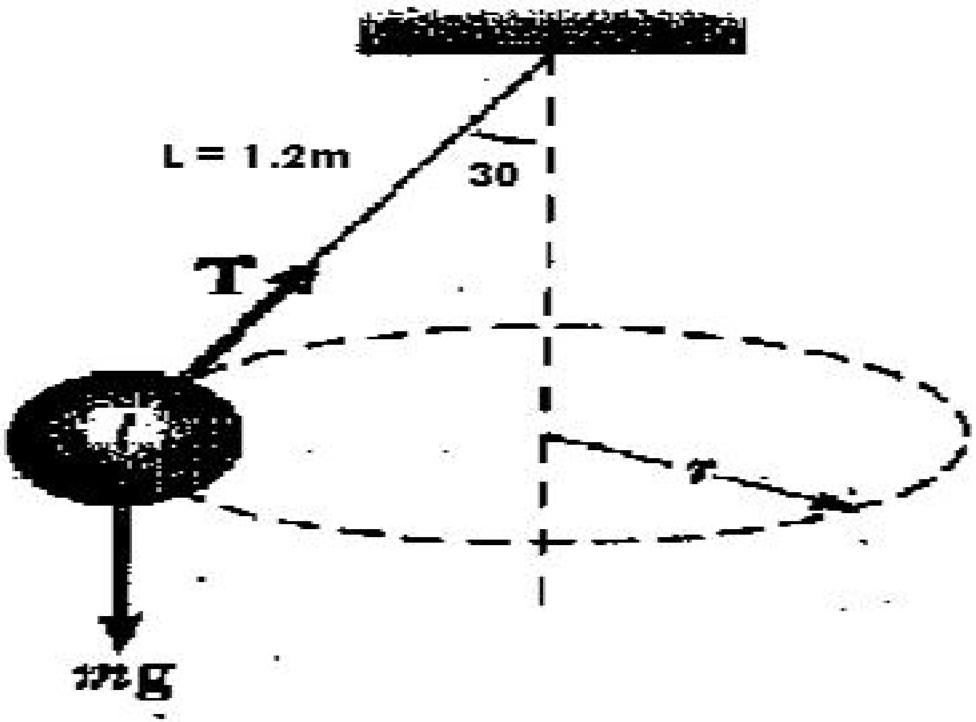


$$\sin\theta = \frac{opp.}{hypo.} = \frac{r}{l}$$

$$\boxed{v = \sqrt{L\,g\sin\theta\tan\theta}}$$

Example

A 0.15 kg ball attached to a string which is 1.2 m in length moves in a horizontal circle. The string makes an angle of 30° with the vertical. Find the tension in the string & the speed of the ball.



component-y : $T\cos 30° = mg$

$$T = \frac{mg}{\cos 30°}$$

$$= \frac{0.15(9.81)}{\cos 30°}$$

$$\boxed{T = 1.70\ \mathrm{N}}$$

component - x :

$$T\sin\theta = \frac{mv^2}{r}$$

$$v=\sqrt{\frac{r\,T\sin\theta}{m}}$$

$$v=\sqrt{\frac{(L\sin\theta)\,T\sin\theta}{m}}$$

$$=\sqrt{\frac{(1.2\sin 30)1.7\sin 30}{0.15}}$$

$$\boxed{v=1.84\,m\,s^{-1}}$$

Example

Figure 6.10 shows a conical pendulum with a bob of mass 80.0 kg on a 10.0 m long string making an angle of 5.00° to the vertical.

a. Sketch a free body diagram of the bob.

b. Determine

i. the tension in the string,

ii. the speed and the period of the bob,

iii. the radial acceleration of the bob.

(Given g =9.81 m s^{-2})

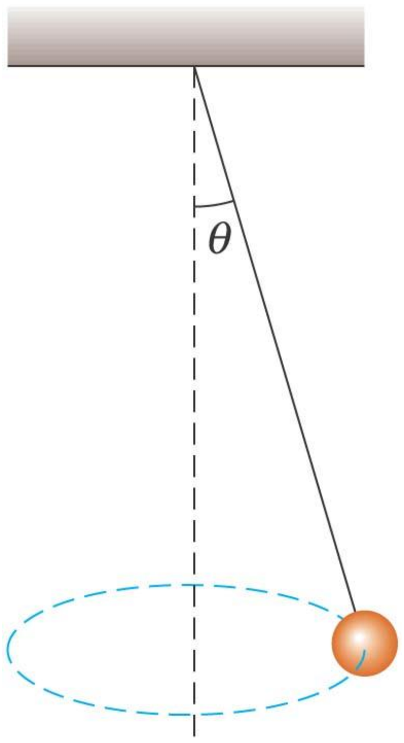
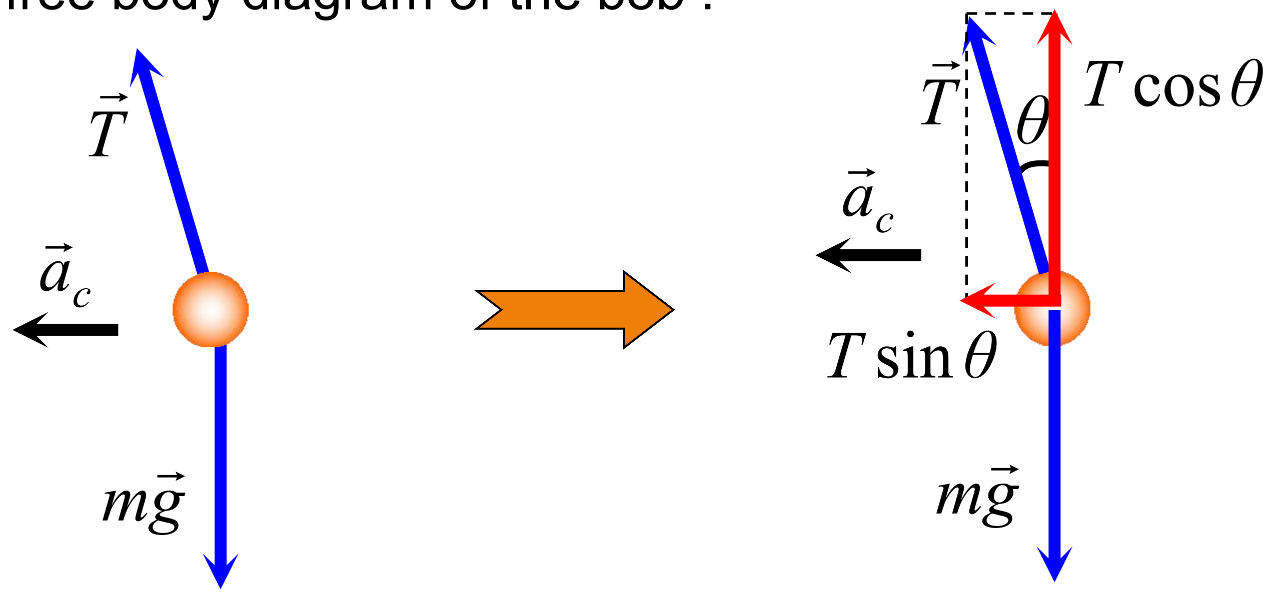


Figure 6.10

Solution : $m = 80.0\ \mathrm{kg};\ l = 10.0\ \mathrm{m};\ \theta = 5.00^\circ$

a. The free body diagram of the bob :

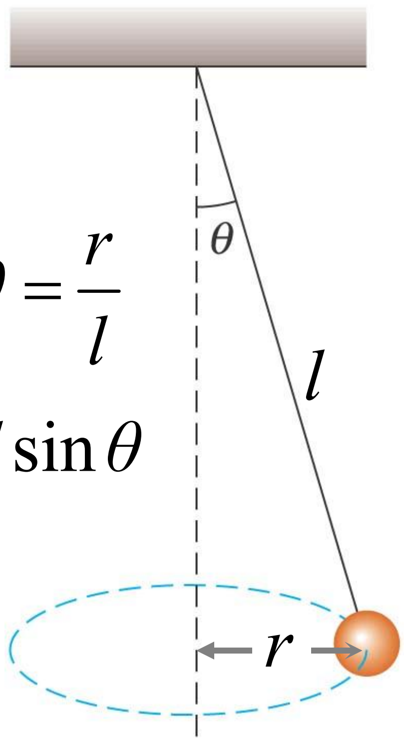


b. i. From the diagram,

$$\sum F_y = 0$$
$$T\cos\theta = mg$$
$$T\cos 5.00^\circ = (80.0)9.81 \Rightarrow T = 788\ \mathrm{N}$$

Solution : $m = 80.0\,\text{kg};\, l = 10.0\,\text{m};\, \theta = 5.00^\circ$

b. ii.



$$\sin\theta = \frac{r}{l}$$

$$r = l\sin\theta$$

The centripetal force is contributed by the horizontal component of the tension.

$$\sum F_x = F_c$$

$$T\sin\theta = \frac{mv^2}{r}$$

$$T\sin\theta = \frac{mv^2}{l\sin\theta}$$

$$v = \sqrt{\frac{Tl\sin^2\theta}{m}}$$

$$v = \sqrt{\frac{(788)(10.0)(\sin 5.00^\circ)^2}{80.0}}$$

$$v = 0.865\,\text{m s}^{-1}$$

Solution : $m = 80.0\,\text{kg};\, l = 10.0\,\text{m};\, \theta = 5.00^\circ$

b. ii. and the period of the bob is given by

$$v = \frac{2\pi r}{T}$$

$$v = \frac{2\pi(l\sin\theta)}{T} \Rightarrow 0.865 = \frac{2\pi(10.0\sin 5.00^\circ)}{T}$$

$$T = 6.33\,\text{s}$$

iii. From the definition of the radial acceleration, hence

$$a_r = \frac{v^2}{r} \Rightarrow a_r = \frac{v^2}{l\sin\theta}$$

$$a_r = \frac{(0.865)^2}{10.0\sin 5.00^\circ}$$

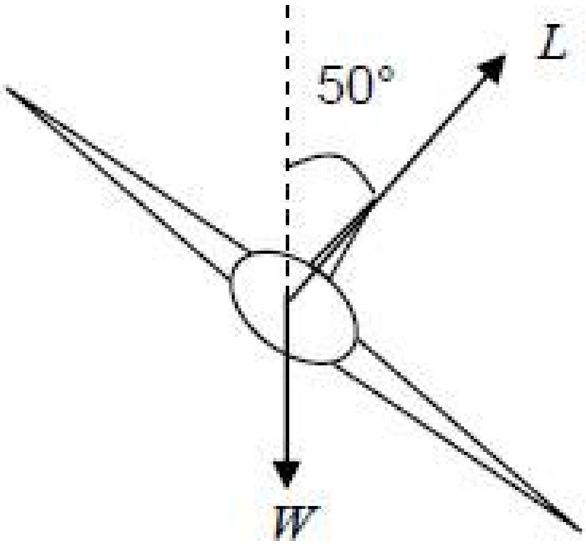
$$a_r = 0.859\,\text{m s}^{-2}$$

(towards the centre of the horizontal circle)

Example

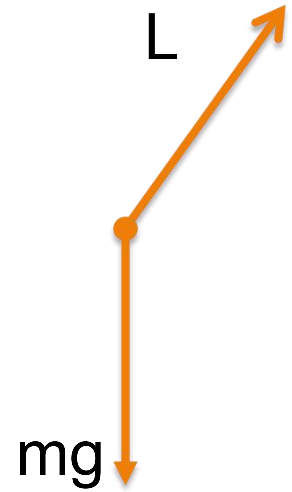
An airplane is turning to the right following a horizontal path with a radius of 25km. The body of the plane is inclined at a 50^o angle to the vertical axis.

a. Sketch a free-body diagram to show the forces acting on the airplane.
b. Determine the speed of the airplane.

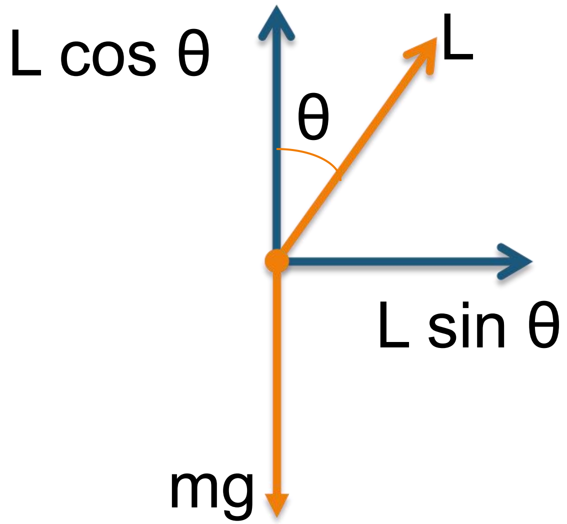


Solution:

a. free-body diagram



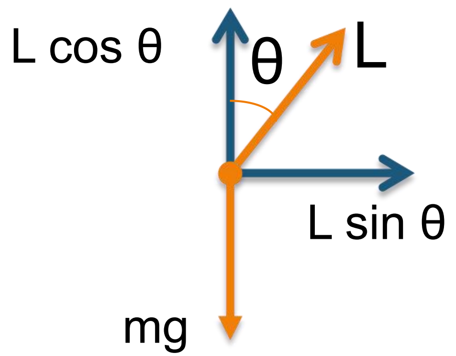
b. Speed of the airplane



$$\text{At equilibrium, } \sum F_y = 0$$

$$L\cos\theta - mg = 0$$

$$L\cos\theta = mg \quad .... (1)$$



$$\sum F_x = ma_c$$

$$L\sin\theta = \frac{mv^2}{r} \quad ....(2)$$

$$(2)\div(1): \quad \frac{\cancel{L}\sin\theta}{\cancel{L}\cos\theta} = \frac{\cancel{m}v^2/r}{\cancel{m}g}$$

$$\tan\theta = \frac{v^2}{rg} \Rightarrow v = \sqrt{rg\tan\theta}$$

$$v = \sqrt{\left(25\times10^3\right)(9.81)\tan 50}$$

$$v = 540.6\,ms^{-1} = 1946 \text{ km hr}^{-1}$$

Follow up Exercise

LO 6.1

1. A manufacturer of CD-ROM drives claims that the player can spin the disc as frequently as 1200 revolutions per minute.

   (a) If spinning at this rate, what is the speed of the outer row of data on the disc if this row is located 5.6 cm from the center of the disc?

   (b) What is the acceleration of the outer row of data?

   ANS: (a) **7.04** m s^{-1} ***(b)*** **8.84**$\times 10^2$ m s^{-2}

LO 6.2

2. (a) A 3 kg mass attached to a light string rotates on a horizontal frictionless table. The radius of the circle is 0.8 m and the string can support a mass of 25 kg before breaking. What range of speeds can the mass have before the string breaks?

Follow up Exercise

(b)

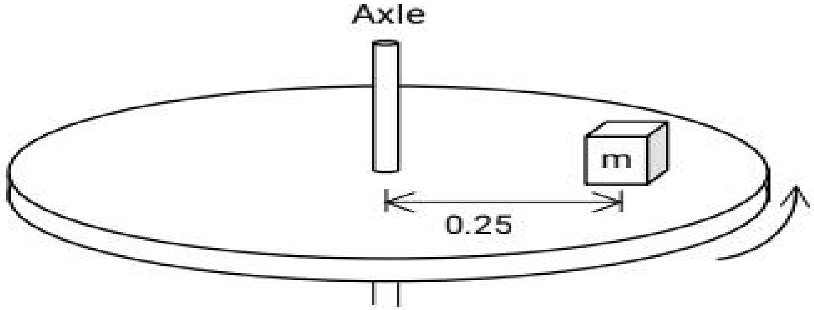


FIGURE 1

An object of mass m is on a horizontal rotating platform as shown in **FIGURE 1**. The mass is located 0.25 m from the axle and makes one revolution every 0.74 s. The friction force needed to keep the mass from sliding is 15 N. What is the object's mass?

ANS: (a) **8.09** m s^{-1} ***(b)*** 0.83 kg

LO 6.2

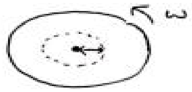
3. A small body of mass 0.50 kg is attached by a 0.75 m long cord to a pin set into the surface of a frictionless table top. The body moves in a circle on the horizontal surface with a speed of 2π m s^{-1}.

(a) What is the magnitude of the radial acceleration of the body?

(b) What is the tension in the cord?

ANS: (a) 52.64 m s^{-1} ***(b)*** 26.32 N

(1)



$r = 5.6\text{ cm}$

$r = 5.6\times10^{-2}\text{ m}$

$\omega = 1200 \frac{\text{rev}}{\text{min}}$

$= 1200 \times \frac{4\pi}{60}$

$\omega = 40\pi \text{ rad s}^{-1}$

a) $v = r\omega$

$v = (5.6\times10^{-2})(40\pi)$

$v = 7.04\text{ ms}^{-1}$

b) $a_c = \frac{v^2}{r}$

$= \frac{(7.04)^2}{(5.6\times10^{-2})}$

$a_c = 884.32\text{ ms}^{-2}$

(2)

$m = 3\text{kg}$ $m = 25\text{kg}$

FBD :

$\Sigma F_x = ma$

$T = ma_c$

$T = \frac{m(v^2)}{r}$

max $T = 245.25\text{ N}$

$\therefore T = \frac{mv^2}{r}$

$245.25 = \frac{(3)v^2}{0.8}$

$v = 8.09\text{ ms}^{-1}$

Before breaking

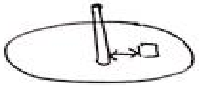
$T_{max} = W$

$T_{max} = mg$

$T_{max} = 25(9.81)$

$T_{max} = 245.25\text{ N}$

(2) b)



$r = 0.25\text{m}$

$T = 0.74\text{ s}$

$\omega = \frac{2\pi}{T}$

$\omega = \frac{2\pi}{0.74}$

$\omega = 8.49\text{ rads}^{-1}$

$f_s = 15\text{ N}$.

FBD :

$\Sigma F_x = ma$

$f_s = ma_c$

$15 = m\frac{v^2}{r}$

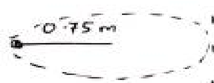
$15 = mr\omega^2$

$15 = m(0.25)(8.49)^2$

$15 = m(18.02)$

$m = 0.83\text{ kg}$

(3)



$m = 0.50\text{ kg}$

$r = 0.75\text{ m}$

$V = 2\pi\text{ ms}^{-1}$

a) (radius) radial acceleration = centripetal acceleration

$a = \frac{v^2}{r} = \frac{(2\pi)^2}{0.75} = 52.64\text{ ms}^{-2}$

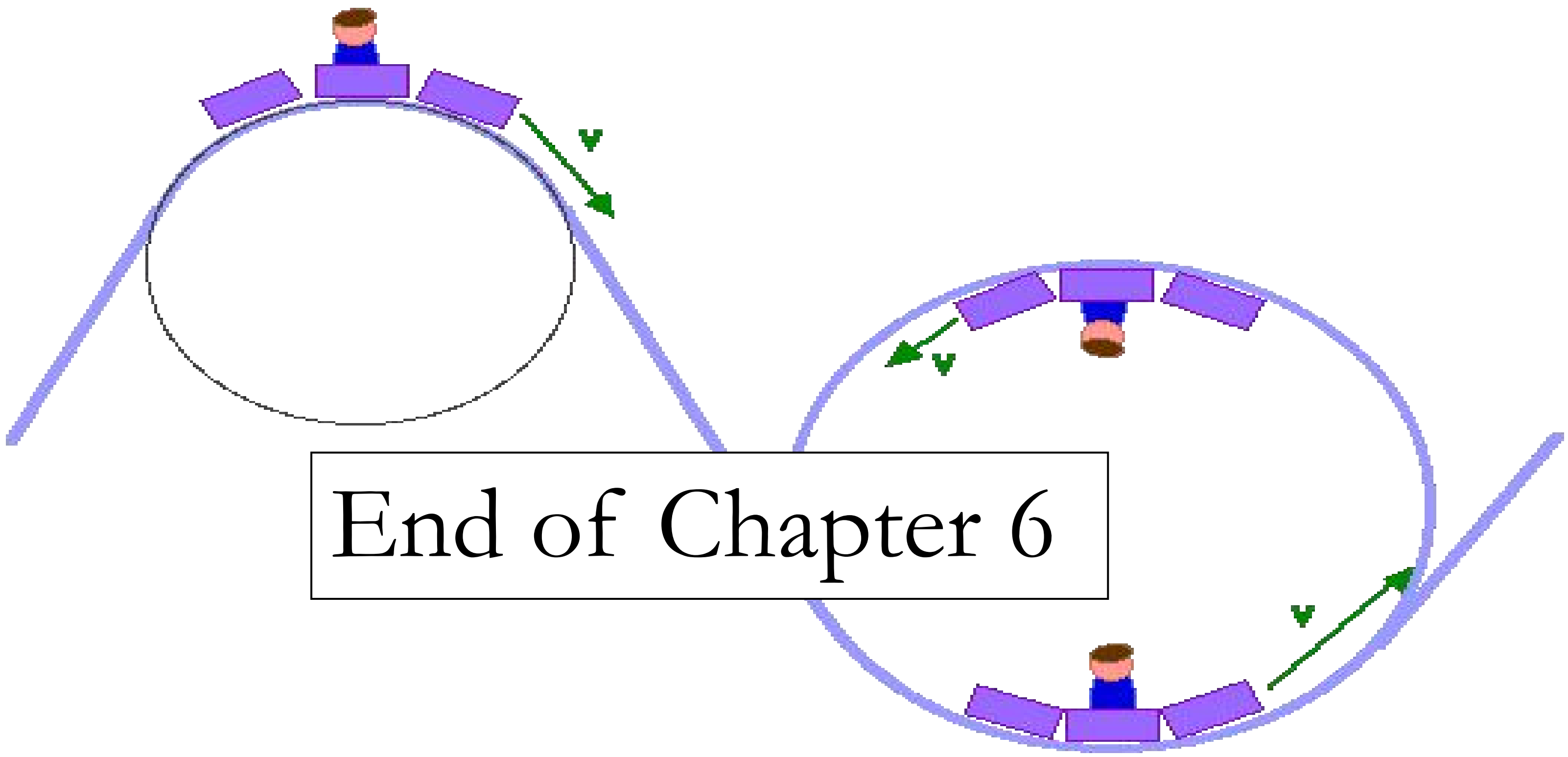
b)

$\Sigma F_x = ma$

$T = ma_c$

$T = (0.5)(52.64)$

$T = 26.32\text{ N}$



End of Chapter 6